

Open Watcom C/C++

Getting Started



Version 2.0

Open Watcom

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For more information please visit <https://github.com/open-watcom/open-watcom-v2> .

Table of Contents

1 Introduction to Open Watcom C/C++	1
1.1 What is in version 2.0 of Open Watcom C/C++?	1
1.2 Technical Support and Services	3
Resources at Your Fingertips	3
Contacting Technical Support	4
Information Technical Support Will Need to Help You	4
Suggested Reading	4
C Programmers	5
C++ Programmers	5
DOS Developers	5
Extended DOS Developers	5
Windows 3.x Developers	5
Windows NT Developers	6
OS/2 Developers	6
Virtual Device Driver Developers	6
2 Installation	7
2.1 Hardware and Software Requirements	7
2.2 The README File	7
2.3 Installing Open Watcom C/C++	7
2.4 Incremental Installation	9
2.5 System Configuration File Modifications	9
2.6 Installation Notes for Windows 3.x	10
2.7 Installation Notes for OS/2	10
3 Hands-on Introduction to Open Watcom C/C++	13
3.1 Outline	13
3.2 The Open Watcom C/C++ Tutorial	14
Defining a Project	14
Adding Multiple Targets	17
Making a Target	17
Making All Targets	17
Executing the Program	18
Smart Editing	19
Debugging the Program	21
Using the Source Browser	23
Correcting an Error	24
Editing a Bitmap	25
Editing Menus	26
Sampling and Profiling an Executable	28
Saving the Project and Terminating the Session	29
3.3 Tutorial Review	30
4 Documentation	31
4.1 Accessing On-line Documentation	32
On-line Documentation under DOS and Linux	32
On-line Documentation under Windows	33
On-line Documentation under OS/2	34
5 Benchmarking Hints	37
6 Release Notes for Open Watcom C/C++ 2.0	39

Table of Contents

6.1 Differences from Open Watcom version 1.9	39
6.2 Differences from Open Watcom version 1.8	40
6.3 Differences from Open Watcom Version 1.7	43
6.4 Changes in 1.8 that may Require Recompilation	46
6.5 Differences from Open Watcom Version 1.6	46
6.6 Changes in 1.7 that may Require Recompilation	48
6.7 Differences from Open Watcom Version 1.5	48
6.8 Differences from Open Watcom Version 1.4	50
6.9 Differences from Open Watcom Version 1.3	53
6.10 Changes in 1.4 that may Require Recompilation	57
6.11 Differences from Open Watcom Version 1.2	57
6.12 Differences from Open Watcom Version 1.1	60
6.13 Differences from Open Watcom Version 1.0	62
6.14 Differences from Version 11.0	63
6.15 Changes in 11.0 that may Require Recompilation	63
6.16 Major Differences from Version 10.6	63
Changes to the C++ Compiler for 11.0	64
Changes to the C Compiler for 11.0	66
Changes to the Code Generator for 11.0	66
Changes to the Compiler Tools for 11.0	66
Changes to the C/C++ Libraries for 11.0	67
Changes to the DOS Graphics Library for 11.0	68
Changes in Microsoft Foundation Classes Support for 11.0	68
Changes in Microsoft Win32 SDK Support for 11.0	68
Changes in Blue Sky's Visual Programmer for 11.0	68
6.17 Changes in 10.6 that may Require Recompilation	68
6.18 Major Differences from Version 10.5	68
Windows 95 Help File Format	69
Changes to the C++ Compiler in 10.6	69
Changes to the C Compiler in 10.6	69
Changes to the C Library in 10.6	70
Changes in Microsoft Foundation Classes Support for 10.6	70
Changes to the Image Editor in 10.6	70
Changes to the Dialog Editor in 10.6	70
Changes to the Resource Editor in 10.6	70
Changes to the Resource Compiler in 10.6	71
6.19 Major Differences from Version 10.0	71
Changes in 10.5 that may Require Recompilation	81
6.20 Major Differences from Version 10.0 LA	81
6.21 Major Differences from Watcom C9.5 /386	81
Items No Longer Supported	82
Changes in 10.0 that may Require Recompilation	82
6.22 Major Differences from Watcom C9.01 /386	83
Changes that may Require Recompilation	83
6.23 Major Differences from Watcom C9.0 /386	84
Command Line Options added to Watcom C9.0 /386	84
6.24 Major Differences from Watcom C8.5 /386	84
Command Line Options added to Watcom C8.5 /386	85
6.25 Major Differences from Watcom C8.0 /386	85
Command Line Options added to Watcom C8.0 /386	86
6.26 Major Differences from Watcom C7.0 /386	86
Protected-mode Compiler and Linker	86

Table of Contents

7 Sybase Open Watcom Public License	87
8 Trouble-Shooting	93
8.1 Win-OS/2 and OS/2 Specific	94

1 Introduction to Open Watcom C/C++

Welcome to the Open Watcom C/C++ 2.0 development system. Open Watcom C/C++ is an Open Source successor to commercial compilers previously marketed by Sybase, Powersoft and originally WATCOM International Corp.

Version 2.0 of Open Watcom C/C++ is a professional, optimizing, multi-platform C and C++ compiler with a comprehensive suite of development tools for developing and debugging both 16-bit and 32-bit applications for DOS, extended DOS, Novell NLMs, 16-bit OS/2, 32-bit OS/2, Windows 3.x, Windows 95/98/Me, Win32s, Windows NT/2000/XP upto Windows 11 (Win32) and Linux.

You should read the entire contents of this booklet, as it contains information on new programs and modifications that have been made since the previous release.

Special NOTE to users of previous versions! See the section entitled "Release Notes for Open Watcom C/C++ 2.0" on page 39 to determine if you need to recompile your application.

1.1 What is in version 2.0 of Open Watcom C/C++?

Version 2.0 incorporates the features professional developers have been demanding:

Open, Multi-target Integrated Development Environment

The IDE allows you to easily edit, compile, link, debug and build applications for 16-bit systems like DOS, OS/2 1.x, and Windows 3.x and 32-bit systems like extended DOS, Novell NLMs, OS/2, Windows 3.x (Win32s), Windows 95/98/Me, Windows NT/2000/XP upto Windows 11 and Linux. Projects can be made up of multiple targets which permit a project to include EXEs and DLLs. The IDE produces makefiles for the project which can be viewed and edited with a text editor. The IDE is hosted under Windows 3.x, Windows 95/98/Me, Windows NT/2000/XP upto Windows 11 and 32-bit OS/2. (There is no IDE on Linux yet).

The Widest Range of Intel x86 Platforms

Host Platforms

- DOS (command line)
- 32-bit OS/2 (IDE and command line)
- Windows 3.x (IDE)
- Windows 95/98/Me (IDE and command line)
- Windows NT/2000/XP upto Windows 11 (IDE and command line)
- Linux (command line)

16-bit Target Platforms

- DOS
- Windows 1 upto Windows 3.x
- OS/2 1.x

32-bit Target Platforms

- Extended DOS
- Win32s
- Windows 95/98/Me
- Windows NT/2000/XP upto Windows 11
- 32-bit OS/2
- Novell NLMs
- Linux 32/64-bit Intel CPU

Cross-Platform Development Tools

The core tools in the package permit cross-platform development that allows developers to exploit the advanced features of many well known 32-bit operating systems, including Windows 95/98/Me, Windows NT/2000/XP upto Windows 11, OS/2 and Linux. Cross-platform support allows you to develop on a host development environment for execution on a different target system.

Multi-Platform Debugger

The new debugger advances developer productivity. New features include redesigned interface, ability to set breakpoints on nested function calls, improved C++ and DLL debugging, reverse execution, and configurable interface. Graphical versions of the debugger are available under Windows 3.x, Windows 95/98/Me, Windows NT/2000/XP upto Windows 11, and 32-bit OS/2. Character versions of the debugger are available under DOS, Windows 3.x, Windows NT/2000/XP upto Windows 11, 32-bit OS/2 and Linux. For VIDEO fans, we have kept the command line compatibility from the original debugger.

Class Browser

The Browser lets you visually navigate the object hierarchies, functions, variable types, and constants of your C/C++ application.

Performance Analysis

The Open Watcom Execution Sampler and Open Watcom Execution Profiler are performance analysis tools that locate heavily used sections of code so that you may focus your efforts on these areas and improve your application's performance.

Editor The Open Watcom Editor is a context sensitive source editor, integrated into the Windows 3.x, Windows 95/98/Me and Windows NT/2000/XP (upto Wondows 11) version of the IDE.

Graphical Development Tools

Open Watcom C/C++ includes a suite of graphical development tools to aid development of Windows 3.x, Windows 95/98/Me and Windows NT/2000/XP upto Windows 11 applications. The development tools include:

Resource Editors Enable you to create resources for your 16-bit and 32-bit Windows applications. For 32-bit OS/2 PM development, Open Watcom C/C++ interoperates with IBM's OS/2 Developer's Toolkit (available from IBM). These tools have been seamlessly integrated into the IDE. The resource compiler allows you to incorporate these resources into your application.

Resource Compiler	Produces a compiled resource file from a source file.
Zoom	Magnifies selected sections of your screen.
Heap Walker	Displays memory usage for testing and debugging purposes.
Spy	Monitors messages passed between your application and Windows.
DDESpy	Monitors all DDE activity occurring in the system.
Dr. Watcom	Enables you to debug your program by examining both the program and the system after an exception occurs; monitors native applications running under Windows 3.x, Windows 95/98/Me or Windows NT/2000/XP upto Windows 11.

Assembler

An assembler is included in the package. It is compatible with a subset of the Microsoft macro assembler (MASM).

C++ Class Libraries

Open Watcom C/C++ includes container and stream class libraries.

Royalty-free 32-bit DOS Extender

Open Watcom C/C++ includes the DOS/4GW 32-bit DOS extender by Tenberry Software with royalty-free run-time and virtual memory support up to 32MB.

Support for wide range of DOS Extenders

Open Watcom C/C++ allows you to develop and debug applications based on the following DOS extender technology: CauseWay DOS Extender, Tenberry Software's DOS/4G and Phar Lap's TNT DOS Extender. You can also develop applications using DOS/32A and FlashTek's DOS Extender but, currently, there is no support for debugging these applications.

Sample programs and applications

Open Watcom C/C++ includes a large set of sample applications to demonstrate the integrated development environment.

1.2 Technical Support and Services

We are committed to ensuring that our products perform as they were designed. Although a significant amount of testing has gone into this product, you may encounter errors in the software or documentation. Technical support is provided on an informal basis through the "Open Watcom Discord Channel" at <https://discord.com/channels/922934435744206908> or by "GitHub issues" at <https://github.com/open-watcom/open-watcom-v2/issues>. Please visit <https://github.com/open-watcom/open-watcom-v2> for more information.

Resources at Your Fingertips

Open Watcom C/C++ contains many resources to help you find answers to your questions. The documentation is the first place to start. With each release of the product, we update the manuals to answer the most frequently asked questions. Most of this information is also accessible through on-line help.

The "README" file in the main product directory contains up-to-date information that recently became available.

Answers to frequently asked questions are available on the "Open Watcom Web site" at <https://github.com/open-watcom/open-watcom-v2>.

Contacting Technical Support

Our technical support is available to help resolve technical defects in the software. Note that all support is currently informal and free. The following are ways to contact technical support.

Open Watcom Discord Channel The easiest way to get support is through the "Open Watcom Discord Channel" at <https://discord.com/channels/922934435744206908>.

World Wide Web You can also submit bug reports or enhancement requests through the bug tracking on "GitHub issues" at <https://github.com/open-watcom/open-watcom-v2/issues>.

Information Technical Support Will Need to Help You

The more information you can provide to technical support, the faster they can help you solve your problem. A detailed description of the problem, short sample program, and a summary of steps to duplicate the problem (including compiler and linker options) are essential. Concise problem reports allow technical support to quickly pinpoint the problem and offer a resolution. Here is a list of information that will help technical support solve the problem:

Contact information

We would like your name, as well as your email address, where you can be reached.

Product information

Please tell us the product name and exact version number.

Hardware configuration

Please tell us what type of processor you are using (e.g., 2.2GHz Intel Pentium 4), how much memory is present, what kind of graphics adapter you are using, and how much memory it has.

Software configuration

Please tell us what operating system and version you are using.

Concise problem report with short sample program

Please provide a complete description of the problem and the steps to reproduce it. A small, self-contained program example with compile and link options is ideal.

Suggested Reading

There are a number of good books and references that can help you answer your questions. Following is an old list of some of the books and documents we feel might be helpful. This is by no means an exhaustive list. Contact your local bookstore for additional information.

C Programmers

The C Programming Language, 2nd Edition

Brian W. Kernighan and Dennis M. Ritchie; Prentice Hall, 1988.

C DiskTutor

L. John Ribar; Osborne McGraw-Hill, 1992.

C++ Programmers

C++ Primer, 2nd Edition

Stanley B. Lippman; Addison-Wesley Publishing Company, 1991.

Teach Yourself C++ in 21 Days

Jesse Liberty; Sams Publishing, 1994.

DOS Developers

PC Interrupts, Second Edition

Ralf Brown and Jim Kyle; Addison-Wesley Publishing Company, 1994.

Relocatable Object Module Format Specification, V1.1

The Tool Interface Standards (TIS) OMF specification can be obtained from the Open Watcom website. Here is the URL.

<http://www.openwatcom.org/ftp/devel/docs/omf.pdf>

This file contains a PDF version of the TIS OMF V1.1 specification.

Extended DOS Developers

Extending DOS—A Programmer's Guide to Protected-Mode DOS, 2nd Edition

Ray Duncan, et al; Addison-Wesley Publishing Company, 1992.

DOS Protected-Mode Interface (DPMI) Specification

The DPMI 1.0 specification can be obtained from the Open Watcom website <https://open-watcom.github.io/open-watcom-1.9/devel/dpmi10.pdf>. This file contains a PDF version of the DPMI 1.0 specification.

Alternatively the DPMI 1.0 specification can be obtained at <http://www.delorie.com/djgpp/doc/dpmi/dpmi-specs.tar.gz>. This file contains online HTML version of the DPMI 1.0 specification.

Windows 3.x Developers

Microsoft Windows Programmer's Reference

Microsoft Corporation; Microsoft Press, 1990.

Programming Windows 3.1, Third Edition

Charles Petzold; Microsoft Press, 1992.

Windows Programming Primer Plus

Jim Conger; Waite Group Press, 1992.

Windows NT Developers

Advanced Windows NT

Jeffrey Richter; Microsoft Press. 1994.

Inside Windows NT

Helen Custer; Microsoft Press. 1993.

Microsoft Win32 Programmer's Reference, Volume One

Microsoft Corporation; Microsoft Press, 1993.

OS/2 Developers

The Design of OS/2

H.M. Deitel and M.S. Kogan; Addison-Wesley Publishing Company, 1992.

OS/2 Warp Unleashed, Deluxe Edition

David Moskowitz and David Kerr, et al; Sams Publishing, 1995.

Virtual Device Driver Developers

Writing Windows Virtual Device Drivers

David Thielen and Bryan Woodruff; Addison-Wesley Publishing Company, 1994.

2 Installation

The package contains the following components:

- *Open Watcom C/C++ installer image*
- *This manual*

2.1 Hardware and Software Requirements

Open Watcom C/C++ requires the following minimum configuration:

- IBM PC compatible
- An 80386 or higher processor
- 8 MB of memory
- Hard disk with enough space available to install the components you require.

In addition to the above requirements, you need one of the following operating systems:

- DOS version 5.0 or higher
- Microsoft Windows version 3.1 running in enhanced mode
- Microsoft Windows 95 or higher
- Microsoft Windows NT version 3.1 or higher
- IBM OS/2 2.1 or higher
- Linux 32/64-bit with Kernel 3.x or higher

2.2 The README File

Before you install Open Watcom C/C++, you should read the contents of the "README" file which is stored in the root directory of the CD-ROM. It contains valuable, up-to-date information concerning this product.

2.3 Installing Open Watcom C/C++

The installation program in this version has been completely redesigned with several new "smart" features. If you have installed a previous version of Open Watcom C/C++ then you should install Open Watcom C/C++ 2.0 into the same path (except for the reason described in the following paragraph). It will examine a previous installation to determine what features were previously installed. It will use this information to establish default settings for the installation that you are about to attempt. Of course, you can add or remove features as you progress through the installation steps.

If you are installing only one of the Open Watcom C/C++ or Open Watcom FORTRAN 77 products and you have an older version of the other product, we do NOT recommend that you install the new product into the same directory as the old product. The Open Watcom C/C++ and Open Watcom FORTRAN 77 products are compatible at the same version number. However, the Open Watcom C/C++ and Open Watcom FORTRAN 77 products are usually NOT compatible across different version numbers. If this is the case, care must be exercised when switching between use of the two products. Environment variables such as **PATH** and **WATCOM** must be modified and/or corrected. System files such as `CONFIG.SYS` and `SYSTEM.INI` must be modified and/or corrected.

If you are installing both Open Watcom C/C++ 2.0 and Open Watcom FORTRAN 77 2.0, we recommend that you install both products under the same directory. This will eliminate duplication of files and, as a result, reduce the total required disk space. The two products share the use of certain environment variables which point to the installation directory. If separate installation directories are used, problems will arise.

When you install Open Watcom C/C++ and Open Watcom FORTRAN 77 in the same directory, you should not deselect any options when running the second installation; otherwise the second product's install may remove files that were installed (and are required) by the first product's install. This isn't an issue if you only have one of Open Watcom C/C++ or Open Watcom FORTRAN 77. The problem is that Open Watcom C/C++ and Open Watcom FORTRAN 77 don't know about the installation options you have selected for each other's product.

If you wish to create a backup of your previous version, please do so before installing Open Watcom C/C++ 2.0.

If you decide to install Open Watcom C/C++ 2.0 into a different directory than the previously installed version, you will have to manually edit system files (e.g., `CONFIG.SYS`, `AUTOEXEC.BAT`, `SYSTEM.INI`) after the installation process is complete to remove the old version from various environment variables (e.g., `PATH`, `DEVICE=`). This is necessary since the path to the new version will appear after the path to the old version. To avoid this extra work, we recommend installing the new version into the same path as the old version.

As an example, here are a few of the environment variables and "RUN" directives that are modified/added to the OS/2 `CONFIG.SYS` file. You should make sure that all references to the older version of the software are removed.

Example:

```
LIBPATH=...;D:\WATCOM\BINP\DLL;...
SET PATH=...;D:\WATCOM\BINP;D:\WATCOM\BINW;...
SET HELP=...;D:\WATCOM\BINP\HELP;...
SET BOOKSHELF=...;D:\WATCOM\BINP\HELP;...
SET INCLUDE=...;D:\WATCOM\H\OS2;D:\WATCOM\H;
SET WATCOM=D:\WATCOM
SET EDPATH=D:\WATCOM\EDDAT
RUN=D:\WATCOM\BINP\NMPBIND.EXE
```

You may wish to run Open Watcom C/C++ under more than one operating system on the same personal computer. For every operating system that you use, simply start up the operating system and run the corresponding install procedure.

If you run the Windows 3.x installation procedure, you do not need to run the DOS installation procedure also.

If you plan to use Win-OS/2 as a development platform under OS/2, you must run the Windows 3.1 install program (selecting Windows 3.1 host support).

Place the CD-ROM disk in your CD-ROM drive. Select one of the following procedures depending on the host operating system that you are currently running. Below, substitute the CD-ROM drive specification for "x:".

DOS Enter the following command:

```
x:\setup
```

Windows 3.x Start Windows 3.x and choose Run from the File menu of the Program Manager. Enter the following command:

```
x:\setup
```

Windows 95/98/Me Choose Run from the Start menu and enter the following command:

```
x:\setup
```

Windows NT/2000/XP Log on to an account that is a member of the "Administrator" group so that you have sufficient rights to modify the system environment. Choose Run from the File menu of the Program Manager. Enter the following command:

```
x:\setup
```

OS/2 Start an OS/2 session and enter the following command:

```
x:\install
```

2.4 Incremental Installation

You may wish to install Open Watcom C/C++, and subsequently install features that you omitted in the first install. You can also remove features that you no longer wish to have installed. You can achieve this as follows:

1. Start the installation program.
2. Select any new features that you wish to install.
3. Deselect any features that you wish to remove.
4. Re-run the installation program for each host operating system that you use.

2.5 System Configuration File Modifications

The install program makes changes to your operating system startup files to allow Open Watcom C/C++ to run. We strongly recommend that you allow the install program to modify your system configuration files for you, but you may do it by hand. The changes required may be found in any of the following files which have been placed in the root of the installation directory:

CONFIG.NEW Changes required for CONFIG.SYS (DOS, Windows, Windows 95/98/Me, OS/2)

AUTOEXEC.NEW Changes required for AUTOEXEC.BAT (DOS, Windows, Windows 95/98/Me, OS/2)

2.6 Installation Notes for Windows 3.x

1. When you use the Integrated Development Environment under Windows 3.x, it is important that the IDE's batch server program be able to run in the background. Therefore, make sure that the "Exclusive in Foreground" checkbox is NOT checked in the "Scheduling" options of "386 Enhanced" in the "Control Panel".

2. When you use the Integrated Development Environment under Windows 3.x, the line

`OverlappedIO=ON`

in your "SYSTEM.INI" file can cause problems. This controls (disables) the queuing of DiskIO and makes some changes between DOS box timings to allow some processes to finish.

3. When you use the Integrated Development Environment under Windows 3.x, it is important that the line

`NoEMMDriver=ON`

not appear in your "SYSTEM.INI" file. It will prevent a link from succeeding in the IDE..

4. When you use the Integrated Development Environment under Windows 3.x on the NEC PC-9800 series, it is important that the line

`InDOSPolling=TRUE`

not appear in your "SYSTEM.INI" file. It will prevent a make from succeeding in the IDE.

5. Central Point Software's anti-virus programs (VDEFEND, VSAFE, VWATCH) conflict with the Integrated Development Environment under Windows 3.x.
6. The Program Information File "BATCHBOX.PIF" is used by the Integrated Development Environment (IDE) to start up a background batch server for compiling, linking, etc. The PIF references "COMMAND.COM". If you are using a substitute for "COMMAND.COM" such as "4DOS.COM" then you must modify the PIF accordingly using a PIF editor.

2.7 Installation Notes for OS/2

1. The Integrated Development Environment (IDE) uses the IBM OS/2 Enhanced System Editor (EPM) for editing text files. You must ensure that EPM is installed in your OS/2 system if you are planning to use the IDE. You can selectively install the Enhanced Editor by running the OS/2 Setup and Installation program (Selective Install) and choosing "Enhanced Editor" from the "Tools and Games" detail page.
2. On some systems with limited memory that use the UNDELETE feature of OS/2, compile times may be slow because OS/2 is saving copies of compiler temporary files. You may start the BATSERV process using the OS/2 STARTUP .CMD file with **DELDIR** turned off as illustrated below.


```
SET OLD_DEL_DIR=%DELDIR%
SET DELDIR=
DETACH C:\WATCOM\BINP\BATSERV.EXE
SET DELDIR=%OLD_DEL_DIR%
SET OLD_DEL_DIR=
```

3. If you plan to use the Named Pipe Remote Debugging support of the Open Watcom Debugger then the NMPSEV.EXE. program must be running. It may be started during OS/2 initialization via a "RUN=" statement in your CONFIG.SYS file or manually as needed through the DETACH command.

3 Hands-on Introduction to Open Watcom C/C++

Let's get started and introduce some of the tools that are in Open Watcom C/C++. The purpose of this chapter is to briefly test out the new graphical tools in Open Watcom C/C++ using an existing application.

In this tutorial, we will take an existing set of C++ source files, create a project in our integrated development environment, and perform the following tasks:

- Add multiple targets
- Make a target
- Make all targets
- Execute the program
- Debug the program
- Use the Browser
- Correct errors
- Sample and profile the executable
- Save the project
- Terminate the session

3.1 Outline

Open Watcom's Integrated Development Environment (IDE) manages the files and tools that a programmer uses when developing a project. This includes all the source files, include files, libraries, compiler(s), linkers, preprocessors, etc. that one uses.

The IDE has a graphical interface that makes it easy to visualize the make-up of a project. A single IDE session shows a project. If the project consists of a number of components, such as two executables and one library, these are each shown as target windows in the project window. Each target window shows the files that are needed to construct the target and is associated via its filename extension with a rule that describes the construction mechanism. For example, a filename with the extension ".EXE" may be associated with the rule for constructing 32-bit Windows executables, or a filename with the extension ".LIB" may be associated with the rule for constructing static libraries. Different projects can refer to the same target. If they do, the target is shared and can be manipulated via either project, with changes made through one affecting the other.

The IDE itself is a collection of programs that manages the various files and tools used to create the target libraries and executables. It creates makefile(s) from the information in the target descriptions and invokes Open Watcom Make to construct the targets themselves. A configuration file contains built-in knowledge of the Open Watcom compilers, editors, Profiler, and Browser, as well as all their switches.

3.2 The Open Watcom C/C++ Tutorial

This tutorial walks you through the creation and execution of a C/C++ program under Windows. This will give you an understanding of the basic concepts of the IDE and its components, and it will detail the steps involved in project development. The result of this tutorial is a three dimensional drawing of a kitchen which you can manipulate using either the menus or the icons on the toolbar. You can rotate and resize the drawing, as well as adjust the lighting and contrast.

To begin, start the IDE. This is done by double-clicking on the "IDE" icon in the Open Watcom C/C++ window.

A status field at the bottom of the IDE window indicates the function of the icon on the toolbar over which your mouse cursor is currently positioned. If the status area does not show you the function of the icons as you move the mouse cursor over them, check that no item in the menu bar is highlighted (if one is highlighted, press the Alt key).

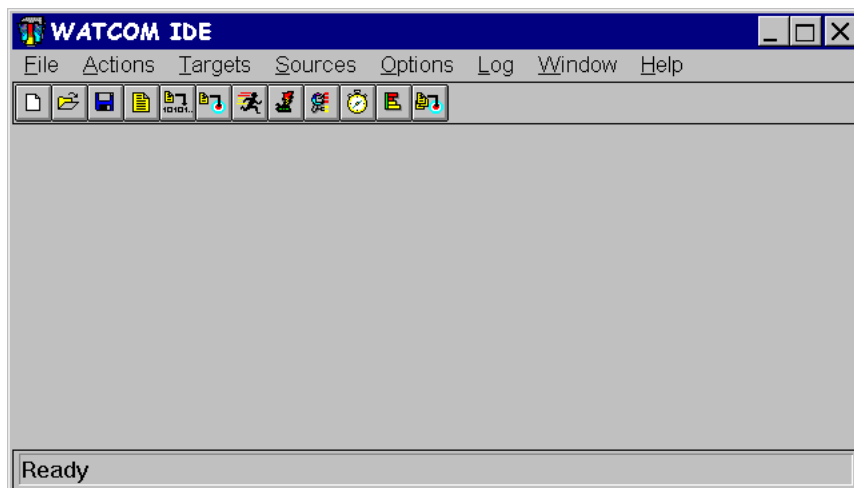


Figure 1. The initial IDE screen

Defining a Project

In this tutorial, you will be creating a new project called `KITCHEN`. Here are the steps required to accomplish this task.

1. Define a new project by pulling down the *File* menu and selecting the *New Project...* item. You can also define a new project by clicking on the "Create a new project" icon on the toolbar.
2. A choice of different sample project directories is available. Assuming that you installed the Open Watcom C/C++ software in the `\WATCOM` directory, you will find the sample project directories in the following directory:

`\WATCOM\SAMPLES\IDE`

For purposes of this tutorial, we recommend that you select one of the following project directories:

WIN	for an example of 16-bit Windows 3.x application development when using Windows 3.x under DOS as a host development system,
WIN386	for an example of 32-bit Windows 3.x application development when using Windows 3.x under DOS as a host development system,
WIN32	for an example of 32-bit Win32 application development when using Windows NT/2000/XP or Windows 95/98/Me as a host development system, and
OS2	for an example of 32-bit OS/2 application development when using 32-bit OS/2 as a host development system.

Thus the `target` that we refer to below should be one of `WIN`, `WIN386`, `WIN32`, or `OS2` depending on your selection. The tutorial uses the `WIN32` example for illustrative purposes. You will find some minor variations from your selected target environment.

When asked for a project name, you can do one of two things:

1. enter the following pathname:

`d: [path] \SAMPLES\IDE\target\KITCHEN`

where `d: [path]` is the drive and path where you installed the Open Watcom software, or

2. use the file browser to select the following directory:

`d: [path] \SAMPLES\IDE\target`

and specify the filename `kitchen`.

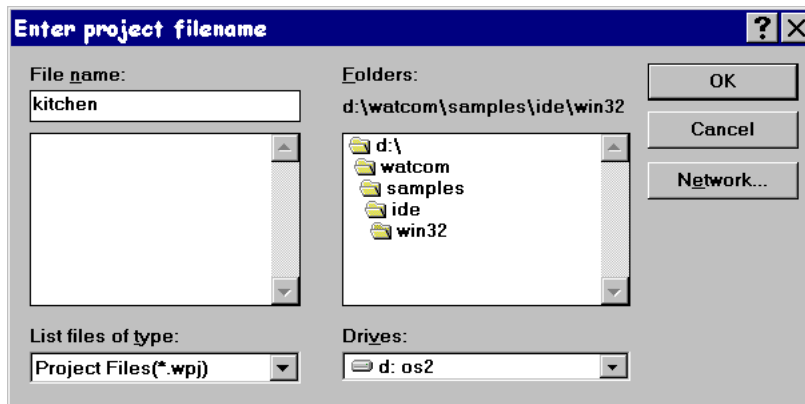


Figure 2. *Creating a new project*

Press the Enter key or click on OK (OPEN).

The project description will be stored in this file and the IDE will set the current working directory to the specified path during your session.

3. You will be prompted for a target name. Since we will be attaching pre-defined targets, just click the *Browse* button when prompted for the target name. Select the "draw" target file (it will be one of `draw16.tgt`, `draw.tgt`, `draw32.tgt`, `drawos2.tgt` depending on your selection of target).

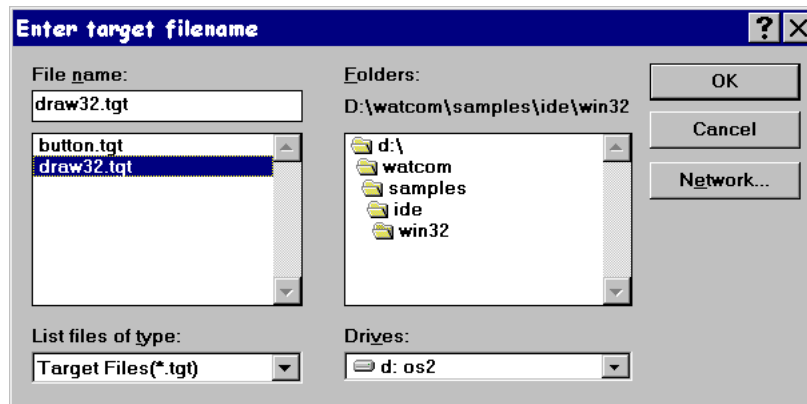


Figure 3. Attaching existing targets

Press the Enter key or click on OK (OPEN).

4. You can ignore the settings displayed for *Target Environment* and *Image Type* since the target definition already exists (we created it for you). The settings are important when you are defining a new target (i.e., one that was not predefined).

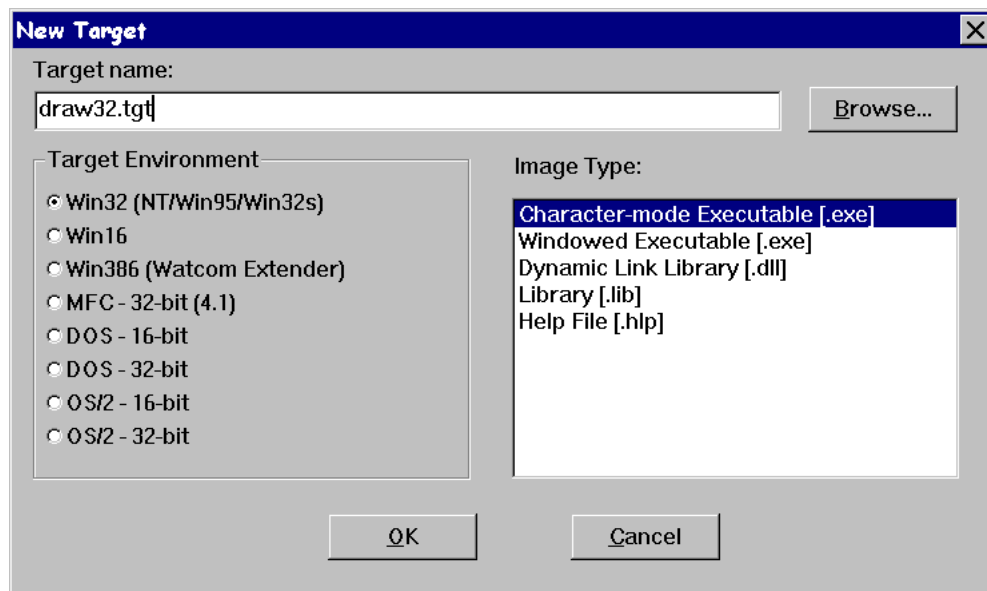


Figure 4. Selecting a target type

Press the Enter key or click on OK.

A target window is created in the project window for the "draw" target. This window contains all of the files associated with the target. You can click on any of the "Folder" icons to hide or un-hide all files with a particular extension. For example, you may wish to un-hide all the files with a .bmp extension by clicking on the folder icon associated with bitmap files.

Adding Multiple Targets

Open Watcom's IDE allows you to have multiple targets in any particular project. Note that targets can be used by multiple projects. To add a new target to the project, do the following.

1. Pull down the *Targets* menu and select the *New Target...* item.
2. Enter `button.tgt` as the target name for the new target to be added to the project. Do not forget to include the `.tgt` extension. It is required when selecting a pre-existing target.
3. You can ignore the settings displayed for *Target Environment* and *Image Type* since the target definition already exists (we created it for you). The settings are important when you are defining a new target (i.e., one that was not predefined).
4. Press the Enter key or click on OK.

A target window is created in the project window for the `button` target.

Making a Target

Open Watcom's IDE will automatically generate the sequence of steps required to build or "make" each of the targets in a project. Note that the targets in a project can be made individually or collectively. To make the `button.lib` target, do the following.

1. Click on the window of the target you wish to make. In this case, click on the `button.lib` target window.
2. Pull down the *Targets* menu and select the *Make* item (you can also do this by clicking on the "Make the current target" icon on the toolbar, or by right-clicking on the target and selecting the *Make* item from the pop-up menu).

The IDE will now construct a makefile based on its knowledge of the target and construction rules, and then invoke the make utility to create the target, in this case `button.lib`. The output of this procedure is displayed in the Log window.

Making All Targets

Click the "Make all targets in the project" icon on the toolbar to make all of the targets in the current project. If one target depends on another target, the latter target will be made first. In this tutorial `button.lib` will be made first (there will be nothing to do since it was made previously) and then `draw???.exe`, since `button.lib` is in the list of files associated with `draw???.exe`. In each case, the IDE constructs makefiles based on its knowledge of the target and construction rules. The output of this procedure is displayed in the Log window.

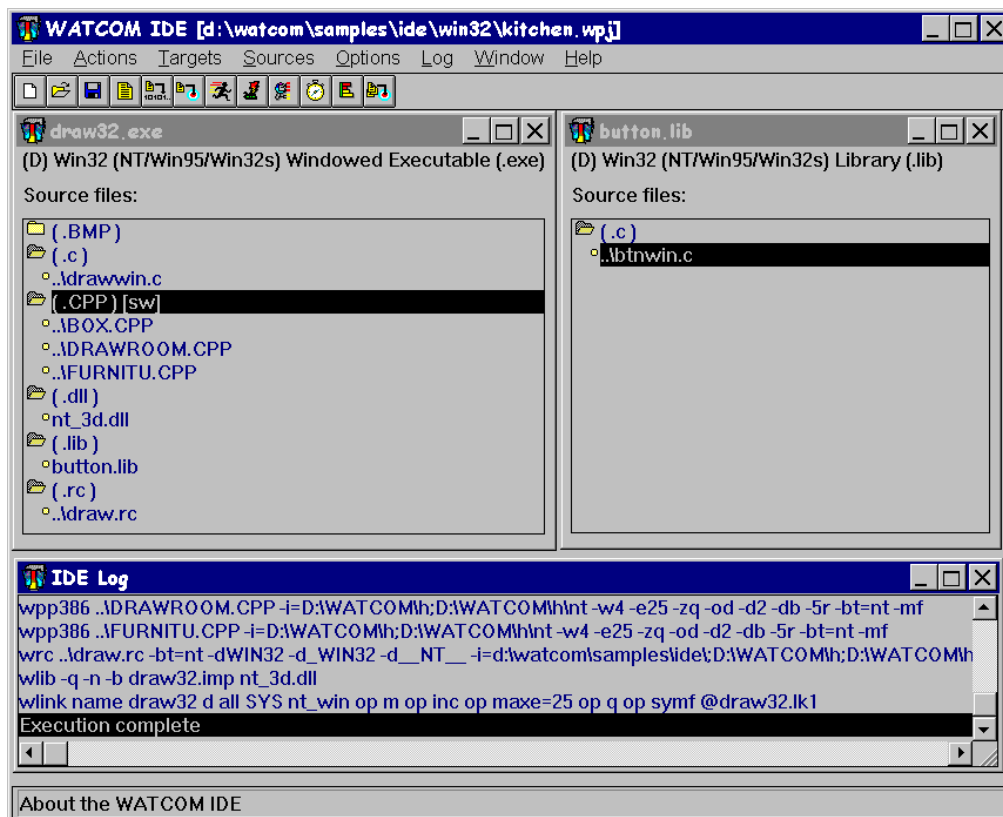


Figure 5. Making one or more targets

Executing the Program

The project should have built without errors, so now you are ready to execute the program you have developed.

Click on the `draw32.exe` target window and simply click on the "Run the current target" icon on the toolbar. You can also do this by selecting *Run* from the *Targets* menu, or right-clicking anywhere on the target window outside of the inner frame and selecting *Run* from the pop-up menu (right-clicking is context sensitive and the pop-up menu that results will vary depending on the area of the window in which you right-click).

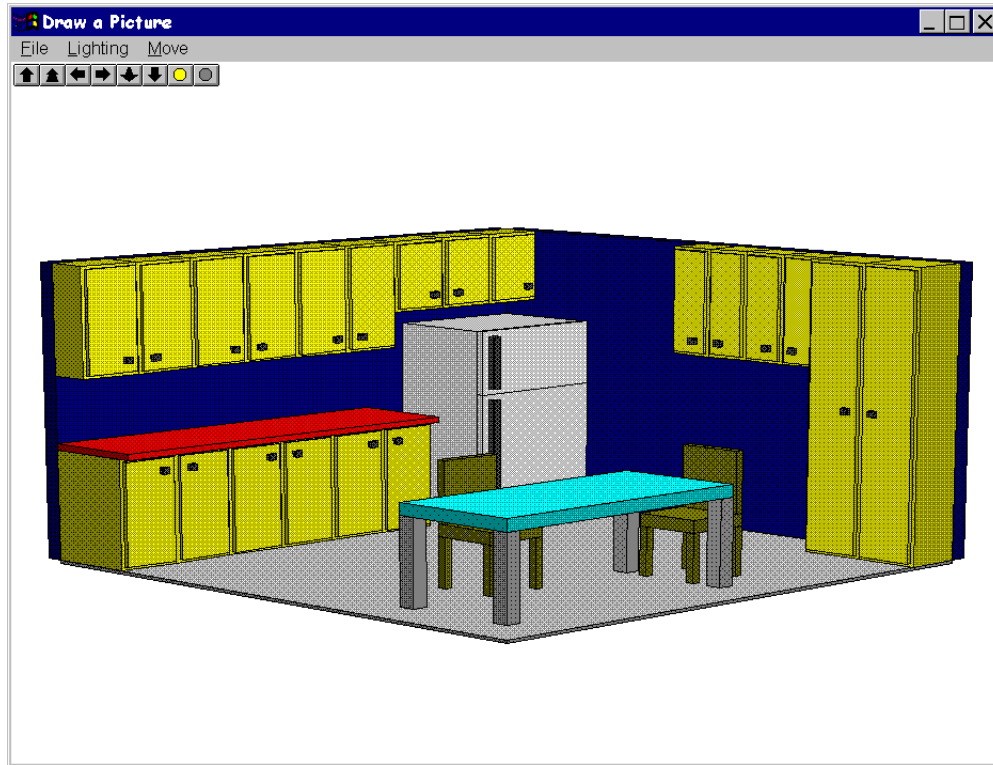


Figure 6. The kitchen demo

The demo you have created is a simple three dimensional drawing of a kitchen. By using either the icons on the toolbar or the menus you can rotate the picture left, right, up, and down, make the picture brighter or dimmer, move the picture closer or farther away, and increase or decrease the amount of contrast (this latter feature is found in the "Lighting" menu). Choose *Exit* from the *File* menu to exit the demo program when you are finished.

Smart Editing

The IDE recognizes the type of file you wish to edit, and invokes the appropriate editor for the task. To edit a file, you either double-click on it or select it and click the "Edit" icon on the toolbar. Files with a `.c`, `.cpp`, `.h`, `.hpp`, `.for`, `.asm` or `.rc` extension are edited with a text editor; files with a `.bmp`, `.ico`, or `.cur` extension are edited with the Image Editor; files with a `.dlg` or `.res` extension are edited with the Resource Editor.

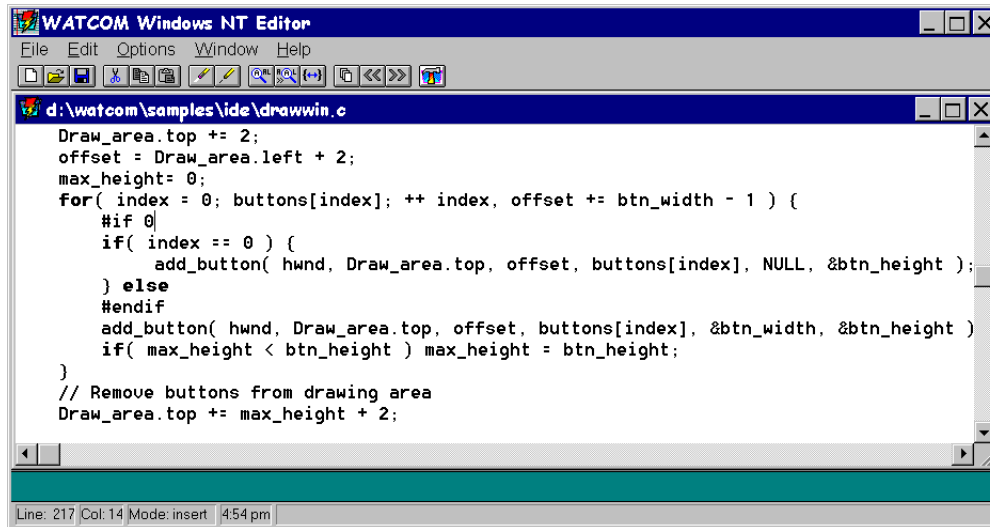


Figure 7. The Open Watcom Editor for Windows

Now we will edit one of the source files and introduce an error into the application.

1. Double-click on the "draw" source file (i.e., draw???.c) to load the source file into the editor.
2. Scroll down to line 227 using the keyboard or mouse. You can also pull down the *Edit* menu, select *Goto Line...*, and enter 227. The Open Watcom Editor makes full use of colors and fonts to achieve syntax highlighting. File templates for C, C++, and FORTRAN files are defined to assist you in distinguishing the components of your code.
3. Pull down the *Options* menu and select the *Colors* item.
4. Click on a color from the palette, drag it to the word `if` on line 218, and release it. All keywords are now displayed in the chosen color. Drag a different color to a comment line (line 225) and all comments will display in that color. Similarly, you can select the *Fonts* item from the *Options* menu, select a font style and size, and drag it to your source file. Close the *Fonts* and *Colors* dialog by double-clicking in the upper left hand corner.
5. You can now save this color and font configuration for all .cpp files by pulling down the *Options* menu, selecting the *General...* item and clicking next to *Save configuration on exit* in the "Features" box. Press Enter or click on OK.
6. Now, to introduce an error into the application, replace the line `#if 0` with `#if 1`.
7. Save your changes by clicking on the "Write the current file to disk" icon or select *Save* from the *File* menu.
8. Return to the IDE (by clicking on it if it is visible on your screen, or by using Alt-Tab), re-make your project, and run it. A fault occurs in your application, so the next step is to track down the problem using the Open Watcom Debugger.

Debugging the Program

To debug a program it must first be compiled to include debugging information. The IDE maintains two sets of switches for each target in a project. These are known as the Development Switches and the Release Switches.

1. Right click on `draw???.c` and select *Source options* from the pop-up menu. Select *C Compiler Switches* from the sub-menu.

By default, your target is placed in development mode with the debugging switches for the compiler and linker set to include debugging information. You can either set the switches in each category manually or you can copy the default Release switches using the CopyRel button. This method of setting switches is especially convenient since you can specify everything from diagnostic, optimization, and code generation switches to special linker switches all without having to memorize a cryptic switch name — you simply click next to the switches you wish to use for a particular item.

2. Scroll through the categories using the >> button until you get to:

6. Debugging Switches

We can see that full debugging information was used in the compile, so just click on *Cancel* to exit this screen.

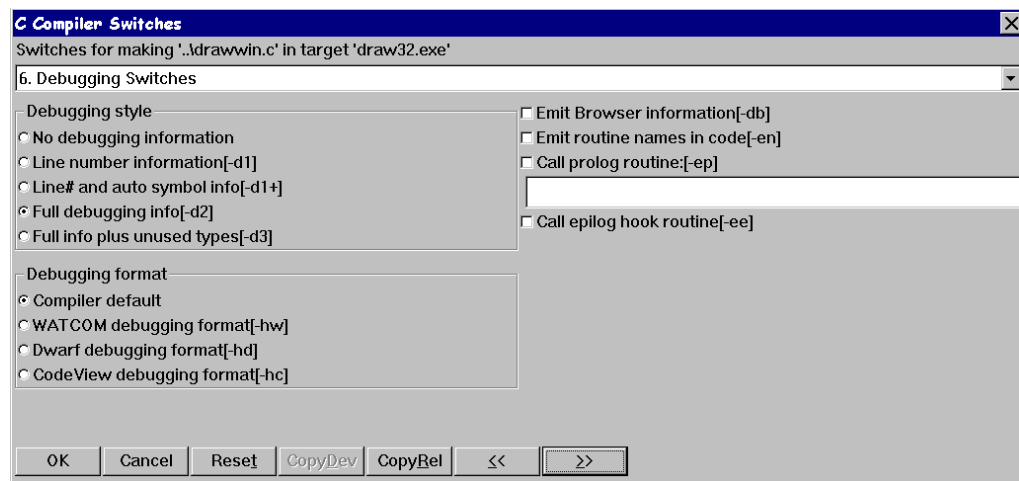


Figure 8. Setting compiler switches

3. To invoke the debugger, pull down the *Targets* menu and choose the *Debug* item or select the "Debug the current target" icon from the toolbar.

The Open Watcom Debugger is designed to be as convenient and intuitive as possible, while at the same time providing a comprehensive and flexible environment for serious debugging. You can configure your environment to display exactly the information you require to be most productive. Among the windows available are source and assembly, modules, functions, calls, threads, images, watches, locals, globals, file variables, registers, 80x87 FPU, stack, I/O ports, memory display, and a log window. You can step through your source using the keys or icons on the toolbar. Execute one line at a time by stepping over calls or stepping into calls, or execute until the current function returns. Right-mouse button functionality gives context-sensitive pop-up menus.

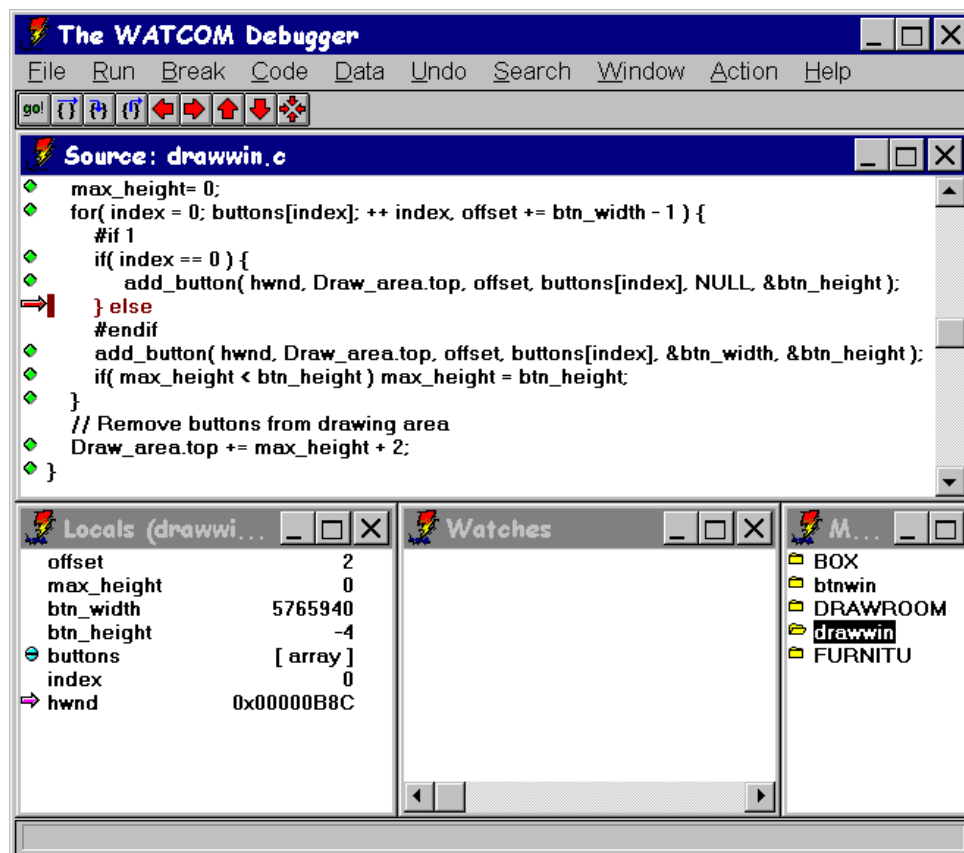


Figure 9. The Open Watcom Debugger

We know that a fault has occurred in `draw??? .exe`, so we will run the application and examine the state of the program when the fault occurs.

1. Click on the "go!" icon on the toolbar to begin execution of the program. The exception occurs and the source window shows the line


```
*pwidth = bitmap.bmWidth + 5;
```

 in the function `button_size` as the last line executed before the exception. Examining the Locals window you will see that `pwidth` is a NULL pointer, hence the exception.
2. We can now move up the call stack by clicking on the "Move up the call stack" icon on the toolbar (red up arrow) to follow the program's execution. On the previous line, we see `button_size` is called from `add_button`. Moving up the call stack again, we see `add_button` is called with NULL as its fifth parameter. An artificial error has been introduced for the purposes of this tutorial. It is located several lines back in the source file.
3. By replacing the line `#if 1` with `#if 0` we can bypass this error. Right-click on the line `#if 1` and select Show, then Line... from the pop-up menus to see the line number which must be corrected, then exit the debugger.
4. Double-click on `draw??? .c` to load the source file into the editor.
5. Scroll down to line 227 using the keyboard or mouse, or pull down the *Edit* menu, select *Goto Line...*, and enter 227.

6. Replace the line `#if 1` with `#if 0` and save your changes by clicking on the "Write the current file to disk" icon or selecting *Save* from the *File* menu.
7. Return to the IDE (by clicking on it if it is visible on your screen, or by using Alt-Tab) and re-make your project.
8. Run your project to see the kitchen demo.

Using the Source Browser

Suppose you wanted to change the color of the tabletop in your application. You can use the Open Watcom Browser to determine the code you will need to change. The Open Watcom Browser provides an easy way to examine the class definitions, member functions, and inheritance trees associated with your C++ code. First, you need to instruct the compiler to emit Open Watcom Browser information.

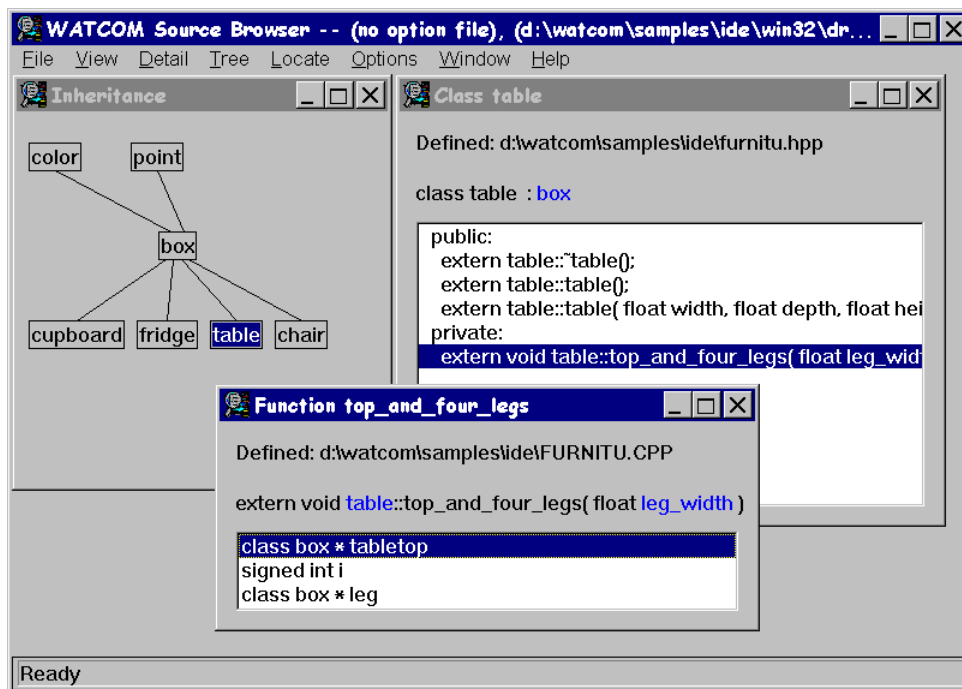


Figure 10. The Open Watcom Browser

1. Right click on `furnitu.cpp`, then select *Source options* from the pop-up menus. Select *C++ Compiler Switches* from the sub-menu.
2. Go to the
 6. Debugging Switches

category by selecting it from the drop-down list box or by scrolling through the categories using the >> button.

3. Select `Emit Browser information [-db]` and click on OK.
4. Click the "Make all targets in the project" icon to re-make the project. The compiler will emit Browser information for `furnitu.cpp` in a file called `draw???.dbr`. Now you are ready to browse the target's source.

5. Pull down the *Targets* menu and select *Browse*, or click the "Browse the current target" icon on the toolbar. The inheritance tree for the target is displayed.
6. To view details on any particular class, double-click on the item for information such as the location of the class definition, the private, public, and protected functions of the class, and the class inheritance. Branches of the inheritance tree can be collapsed and expanded. A variety of tools are available to help you navigate your C++ source. Double-click on the `table` class.
7. Double-click on the function `top_and_four_legs()` to see the details on this function.
8. Select the variable `tabletop`, pull down the *Detail* menu, and select the *Goto Definition...* item. The Editor is invoked, loading the file `furnitu.cpp` which contains the definition of `top_and_four_legs`.
9. Next we will make some changes to `furnitu.cpp` in order to change the color of the `tabletop`. Scroll down to line 143 using the keyboard or mouse, or pull down the *Edit* menu, select *Goto Line...*, and enter 143
10. Replace the line

```
tabletop->rgb(0,255,255);
```

with

```
tabletop->black();
```
11. Save your changes by clicking on the "Write the current file to disk" icon or selecting *Save* from the *File* menu.
12. Shut down the Browser before re-making the project.
13. Return to the IDE (by clicking on it if it is visible on your screen, or by using Alt-Tab).
14. Click the "Make all targets in the project" icon to re-make the project.

Correcting an Error

An error is encountered during the make and error message(s) appear in the log window. Additional information on the error is available by selecting the error, pulling down the *Log* menu and selecting the *Help on Message* item.

1. Double-click on the error message

```
furnitu.cpp (132): Error! E029: (col 15) symbol 'black' has not been  
declared.
```

The offending source file (`furnitu.cpp`) is loaded into the Editor and the cursor is positioned at the line which caused the error. Apparently, `black` has not been defined as a color.

2. Restart the Browser.
3. Double-click on `color` in the Inheritance window to see the member functions of the class `color`. Among our choices are `blue()`, `green()`, and `red()`.

4. Press the Alt-Tab key combination to return to the Editor and replace the line

```
tabletop->black();
```

with

```
tabletop->red();
```

5. Save your changes.
6. Return to the IDE and re-make the project.
7. Run the program to see the changes you have made to the tabletop.

Editing a Bitmap

You can edit bitmaps, icons, or cursors associated with your project using Open Watcom's Image Editor. Double-click on a file with a `.bmp`, `.ico`, or `.cur` extension and the file is loaded into the Image Editor. The editor has many features to design your images, including resizing, rotation, shifting, and a utility to take a "snapshot" of another image and import it.

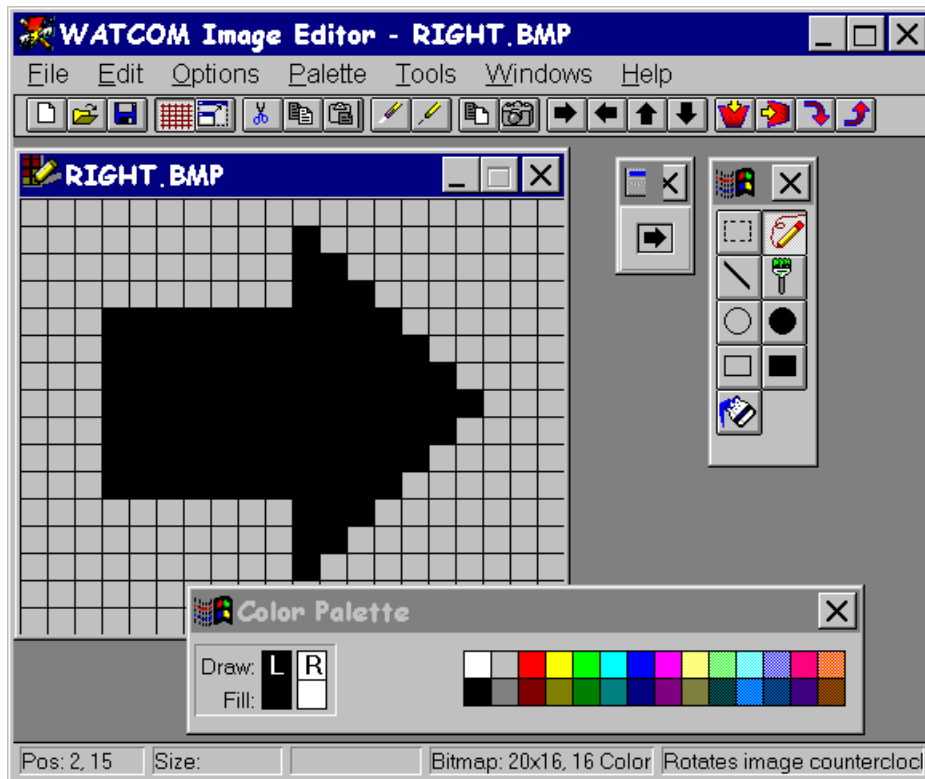


Figure 11. *The Open Watcom Image Editor*

Suppose you wanted to change the color of the right-arrow icon in your application.

1. If the "Folder" icon next to `.bmp` is closed, click on it to restore all the files with a `.bmp` extension to the file list.
2. Scroll the window until the file `right.bmp` is visible.

3. Double-click on `right.bmp` in the `draw???.exe` target window.
4. Select the "Paint Can" icon from the Tool Palette.
5. Select a color from the Color Palette.
6. Click on the arrow.
7. Save your changes using the "Save" icon on the toolbar and exit the Image Editor.
8. Click the "Make all targets in the project" icon to rebuild the project with the change incorporated.

Editing Menus

Next, you will add source files to the list of items that make up `draw???.exe`.

1. Pull down the *Sources* menu and select the *New Source...* item.

Note: You can do this either by choosing from the menu bar or by positioning the mouse over the file list area and clicking the right mouse button. The IDE displays a pop-up menu from which you can choose the desired action.

2. Enter the filename `draw.res` (or `drawos2.res` for OS/2). For OS/2, click OK when you have entered the source file name. For all other systems, click on Add when you have entered the source file name and then click on Close. Now we will remove the `.rc` file from the project so that our changes to the `.res` file will not be overwritten. When an `.rc` file is present, the `.res` file is generated from the `.rc` file.
3. Right click on `draw.rc` (or `drawos2.rc` for OS/2), then select *Remove Source* from the pop-up menu.
4. Double-click on `draw.res` (or `drawos2.res`). The Resource Editor is invoked, displaying all the available resources (in this case, icons, bitmaps, and menus).

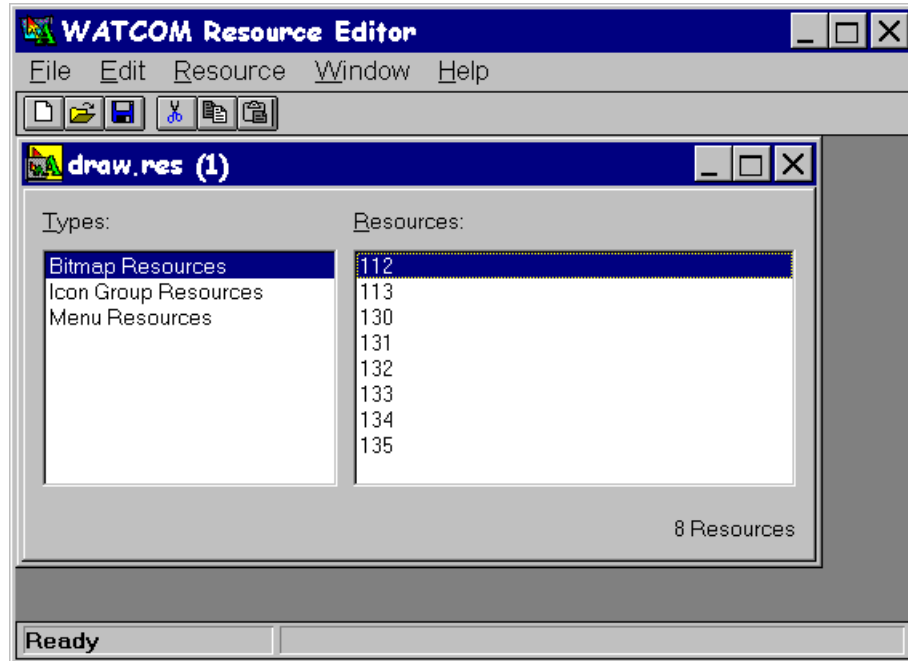


Figure 12. The Open Watcom Resource Editor

5. Click on "Menu Resources".
6. Double-click on "DrawMenu" in the right-hand box. This will bring up the Menu Editor. The Menu Editor displays the menus defined for the resource DrawMenu. You can specify pop-up menus, menu items and sub-items, text, separators, attributes, break styles, and memory flags.

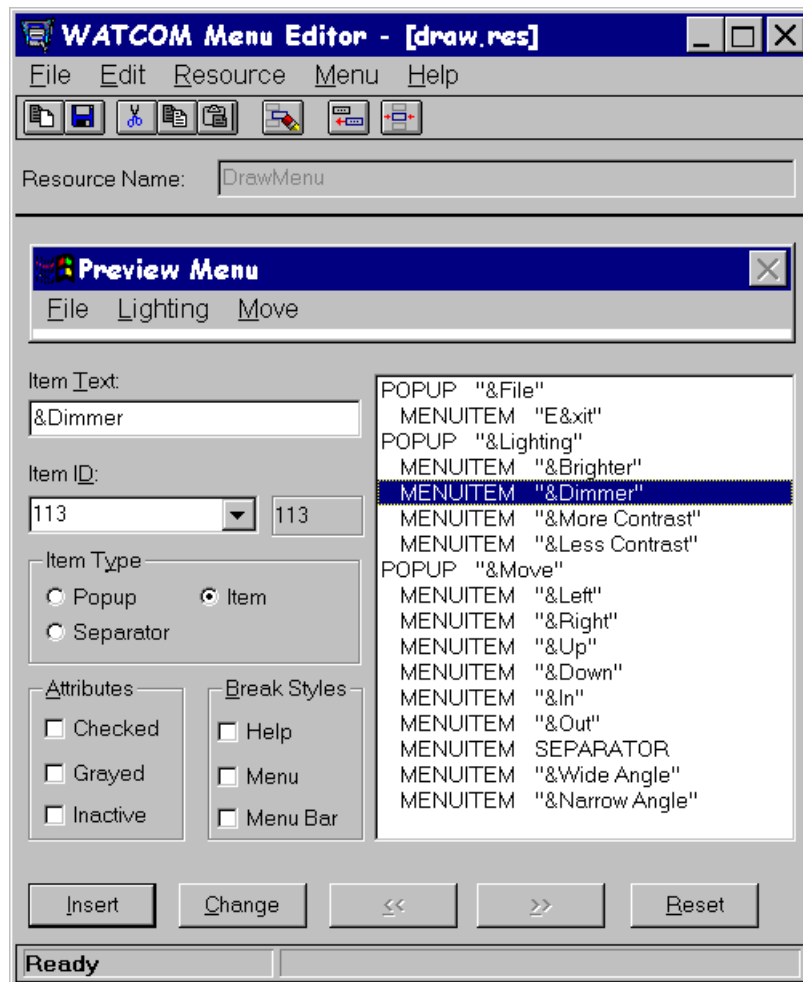


Figure 13. The Open Watcom Menu Editor

7. Click on MENUITEM "&Dimmer" in the item list window.
8. In the "Item Text" window change the item to &Darker and then click on the "Change" button.
9. Select *Update* from the *File* menu or click on the "Update the file with this menu" icon.
10. Exit the Menu Editor.
11. Now, select *Save* from the *File* menu or click on the "Save this file" icon and exit the Resource Editor.
12. Click the "Make all targets in the project" icon to re-make the project.

Sampling and Profiling an Executable

Together, the Open Watcom Execution Sampler and the Open Watcom Execution Profiler allow you to pinpoint the areas of your code that are the most heavily used, indicating possible candidates for performance improvements.

1. Click on the draw???.exe target.

2. Select the *Sample* item from the *Targets* menu item, or click on the "Run and sample the current target" icon from the toolbar. The Open Watcom Execution Sampler is invoked and your application begins to execute.
3. Try rotating and resizing the image a few times. The sampler takes a "snapshot" of the code that is being executed at regular intervals. Exit the application. A samples file with extension `.smp` is created in the current directory. This file is input for the profiler.
4. You are now ready to profile the executable. Do this by selecting *Profile* from the *Targets* menu, or by clicking on the "Profile the current target" icon on the toolbar. The profiler scans the `.smp` file and reports the activity in the various modules of the application. The percentage of time spent in the modules is indicated as an absolute percentage (percent of total samples) and as a relative percentage (percent of samples in the `.exe` image).
5. Double-click on the module or routine names to step down to the exact source being executed when a sample was taken. For more details, you can adjust the sampling rate of the Sampler to get a better picture of your code. To do this, exit the Profiler, pull down the *Targets* menu, and select *Target options*, then *Sample Switches...* from the pop-up menus. Specify a sampling rate such as 2 (for 2 milliseconds), click on OK, then run the Sampler and Profiler again.

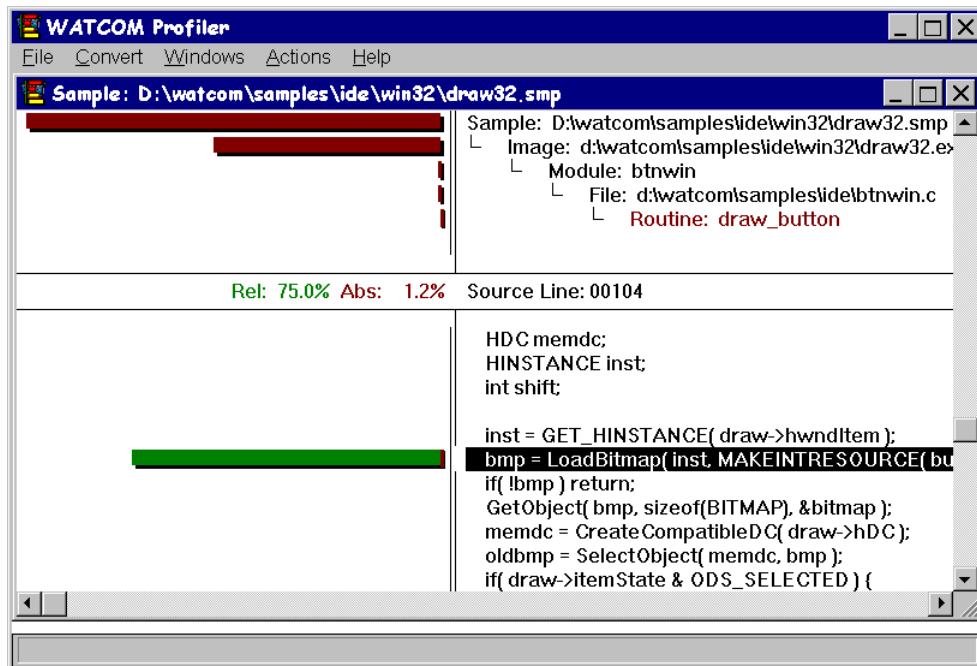


Figure 14. The Open Watcom Execution Profiler

Saving the Project and Terminating the Session

You can now exit the IDE session by selecting *Exit* from the *File* menu. If you have not already saved your project, you will be prompted to do so. Choose "Yes" and the session ends.

3.3 Tutorial Review

In this tutorial, you created a project called `kitchen.wpj`, which was composed of two targets: `draw???.exe` and `button.lib`. You compiled and linked it into an executable program using the WMAKE utility, the Open Watcom C and C++ compilers, and the Open Watcom Linker. You executed it both directly and under the control of the Open Watcom Debugger.. You browsed the source, and made changes using the text and resource editors. Finally, you sampled and profiled the application.

When you saved the project, you created the following permanent files:

- `kitchen.wpj` — describes the screen layout and refers to the target files called `draw???.tgt` and `button.tgt`.
- `draw???.tgt` — describes the target executable `draw???.exe` and all switches required to link it. It also describes the `.c` and `.cpp` files and switches required to compile them.
- `button.tgt` — describes the target library and all switches required to create it. It also describes the `.c` file and the switches used to build the library.

4 Documentation

The following manuals comprise the Open Watcom C/C++ documentation set. When you install the software, portions of the documentation set are provided as on-line help files. Subsequent sections describe how to access this on-line help.

The following describes the titles in the Open Watcom C/C++ documentation set.

Open Watcom C/C++ User's Guide

This manual describes how to use Open Watcom C/C++. It contains an introduction to the compiler and a tutorial section. It also describes compiler options, precompiled header files, libraries, memory models, calling conventions, pragmas, in-line assembly, ROM based applications, and environment variables.

Open Watcom C/C++ Tools User's Guide

This manual describes the command line oriented tools including the compile and link utility, library manager, object file disassembler, far call optimization tool, assembler, patch utility, strip utility, make utility, and touch utility.

Open Watcom Graphical Tools User's Guide

This manual describes Open Watcom's Windows and OS/2 graphical tools including the Integrated Development Environment, Browser, Dr. Watcom, Spy, DDE Spy, Image Editor, Resource Editor, Sampler/Profiler, Resource Compiler, Heap Walker, Zoom, and Editor.

Open Watcom C/C++ Programmer's Guide

This manual includes 5 major sections each of which describes operating system specific development issues. The operating systems covered include extended DOS, OS/2, Windows 3.x, Windows NT/2000/XP, Windows 95/98/Me, 32-bit Windows 3.x (using Open Watcom's Supervisor technology), Novell NLMs and Linux. Topics include creating a sample program, operating system specific error messages, and debugging techniques.

Open Watcom C Language Reference

This manual describes the ISO C programming language and extensions which are supported by Open Watcom C.

Open Watcom C Library Reference

This manual describe the C and graphics libraries supported by Open Watcom C/C++.

Open Watcom C++ Class Library Reference

This manual provides a comprehensive reference to the C++ class libraries provided with Open Watcom C/C++.

Open Watcom Debugger User's Guide

This manual describes the Open Watcom Debugger and discusses advanced debugging techniques.

Open Watcom Linker User's Guide

This manual describes how to use the Open Watcom Linker to generate executables for

target systems such as extended DOS, Windows 3.x, Windows 95/98/Me, Windows NT/2000/XP, OS/2, Novell NLMs and Linux.

4.1 Accessing On-line Documentation

The following sections describe how to access the on-line help that is available for DOS, Windows, OS/2 and Linux.

On-line Documentation under DOS and Linux

The Open Watcom Help program, **WHELP**, may be used under DOS or Linux to access on-line documentation. The Open Watcom Help command line syntax is:

```
WHELP help_file [topic_name]
```

Notes:

1. If *help_file* is specified without an extension then ".ihp" is assumed.
2. The *topic_name* parameter is optional.
3. If *topic_name* is not specified, the default topic is "Table of Contents".
4. If *topic_name* contains spaces then it must be enclosed in quotes.

The following help files are available:

CGUIDE	<i>Open Watcom C/C++ User's Guide</i> (excludes C and C++ Diagnostic Messages appendices which are available as separate help files)
CLIB	<i>Open Watcom C Library Reference</i>
CLR	<i>Open Watcom C Language Reference</i>
CMIX	<i>Open Watcom C/C++ Master Index</i>
CPPLIB	<i>Open Watcom C++ Class Library Reference</i>
LGUIDE	<i>Open Watcom Linker User's Guide</i>
CPGUIDE	<i>Open Watcom C/C++ Programmer's Guide</i>
C_README	<i>Open Watcom C/C++ Getting Started manual</i>
RESCOMP	Documentation for the Open Watcom Resource Compiler (excerpt from the <i>Open Watcom Graphical Tools User's Guide</i>)
CTOOLS	<i>Open Watcom C/C++ Tools User's Guide</i>
WD	<i>Open Watcom Debugger User's Guide</i>

WPROF	Documentation for the Open Watcom Execution Sampler and Open Watcom Execution Profiler (excerpt from the <i>Open Watcom Graphical Tools User's Guide</i>)
WCCERRS	Documentation for the Open Watcom C Diagnostic Messages (excerpt from the <i>Open Watcom C/C++ User's Guide</i>).
WPPERES	Documentation for the Open Watcom C++ Diagnostic Messages (excerpt from the <i>Open Watcom C/C++ User's Guide</i>).

On-line Documentation under Windows

On-line documentation is presented in the form of Windows Help files (".HLP" files). When the software is installed under Windows 3.x, Windows 95/98/Me or Windows NT/2000/XP, a number of program groups are created. You can access the on-line document by opening a program group and double-clicking on a help icon.

Open Watcom C/C++ Group

Getting Started *Open Watcom C/C++ Getting Started*

Open Watcom C/C++ Tools Help Group

Accelerator Editor Help Documentation for the Accelerator Editor (excerpt from the *Open Watcom Graphical Tools User's Guide*)

C Error Messages Documentation for the Open Watcom C Diagnostic Messages (excerpt from the *Open Watcom C/C++ User's Guide*)

C++ Error Messages Documentation for the Open Watcom C++ Diagnostic Messages (excerpt from the *Open Watcom C/C++ User's Guide*)

C Language Reference *Open Watcom C Language Reference*

C Library Reference *Open Watcom C Library Reference*

C++ Library Reference *Open Watcom C++ Class Library Reference*

C/C++ Master Index The master index for all of the Open Watcom C/C++ on-line help

DDE Spy Help Documentation for the DDE Spy utility (excerpt from the *Open Watcom Graphical Tools User's Guide*)

Debugger Help *Open Watcom Debugger User's Guide*

Dialog Editor Help Documentation for the Dialogue Editor (excerpt from the *Open Watcom Graphical Tools User's Guide*)

Dr. Watcom Help Documentation for Dr. Watcom (excerpt from the *Open Watcom Graphical Tools User's Guide*)

Editor Help Documentation for the Open Watcom Editor (excerpt from the *Open Watcom Graphical Tools User's Guide*)

<i>Heap Walker Help</i>	Documentation for the Heap Walker utility (excerpt from the <i>Open Watcom Graphical Tools User's Guide</i>)
<i>IDE Help</i>	Documentation for the Interactive Development Environment (excerpt from the <i>Open Watcom Graphical Tools User's Guide</i>)
<i>Image Editor Help</i>	Documentation for the Image Editor (excerpt from the <i>Open Watcom Graphical Tools User's Guide</i>)
<i>Linker Guide</i>	<i>Open Watcom Linker User's Guide</i>
<i>Menu Editor Help</i>	Documentation for the Menu Editor (excerpt from the <i>Open Watcom Graphical Tools User's Guide</i>)
<i>Profiler Help</i>	Documentation for the Open Watcom Execution Sampler and Open Watcom Execution Profiler (excerpt from the <i>Open Watcom Graphical Tools User's Guide</i>)
<i>Programmer's Guide</i>	<i>Open Watcom C/C++ Programmer's Guide</i>
<i>Resource Compiler Help</i>	Documentation for the Resource Compiler (excerpt from the <i>Open Watcom Graphical Tools User's Guide</i>)
<i>Resource Editor Help</i>	Documentation for the Resource Editor (excerpt from the <i>Open Watcom Graphical Tools User's Guide</i>)
<i>Source Browser Help</i>	Documentation for the Browser (excerpt from the <i>Open Watcom Graphical Tools User's Guide</i>)
<i>Spy Help</i>	Documentation for the Spy utility (excerpt from the <i>Open Watcom Graphical Tools User's Guide</i>)
<i>String Editor Help</i>	Documentation for the String Editor (excerpt from the <i>Open Watcom Graphical Tools User's Guide</i>)
<i>Tools Guide</i>	<i>Open Watcom C/C++ Tools User's Guide</i>
<i>User's Guide</i>	<i>Open Watcom C/C++ User's Guide</i> (excludes C and C++ Diagnostic Messages appendices which are available as separate help files)
<i>Zoom Help</i>	Documentation for the Zoom utility (excerpt from the <i>Open Watcom Graphical Tools User's Guide</i>)

On-line Documentation under OS/2

On-line documentation is presented in the form of OS/2 Information files (".INF" files). When the software is installed under OS/2, the Open Watcom C/C++ folder is created. You can access the on-line document by opening the Open Watcom C/C++ folder and double-clicking on a help icon.

<i>C Error Messages</i>	Documentation for the C Diagnostic Messages (excerpt from the <i>Open Watcom C/C++ User's Guide</i>)
<i>C++ Error Messages</i>	Documentation for the C++ Diagnostic Messages (excerpt from the <i>Open Watcom C/C++ User's Guide</i>)
<i>C Language Reference</i>	<i>Open Watcom C Language Reference</i>
<i>C Library Reference</i>	<i>Open Watcom C Library Reference</i>
<i>C++ Library Reference</i>	<i>Open Watcom C++ Class Library Reference</i>
<i>C/C++ Master Index</i>	The master index for all of the Open Watcom C/C++ on-line help
<i>Debugger Help</i>	<i>Open Watcom Debugger User's Guide</i>
<i>Getting Started</i>	<i>Open Watcom C/C++ Getting Started</i>
<i>IDE Help</i>	Documentation for the Interactive Development Environment (excerpt from the <i>Open Watcom Graphical Tools User's Guide</i>)
<i>Profiler Help</i>	Documentation for the Open Watcom Execution Sampler and Open Watcom Execution Profiler (excerpt from the <i>Open Watcom Graphical Tools User's Guide</i>)
<i>Programmer's Guide</i>	<i>Open Watcom C/C++ Programmer's Guide</i>
<i>Source Browser Help</i>	Documentation for the Open Watcom Browser (excerpt from the <i>Open Watcom Graphical Tools User's Guide</i>)
<i>Tools Guide</i>	<i>Open Watcom C/C++ Tools User's Guide</i>
<i>User's Guide</i>	<i>Open Watcom C/C++ User's Guide</i> (excludes C and C++ Diagnostic Messages appendices which are available as separate help files)

5 Benchmarking Hints

The Open Watcom C/C++ compiler contains many options for controlling the code to be produced. It is impossible to have a certain set of compiler options that will produce the absolute fastest execution times for all possible applications. With that said, we will list the compiler options that we think will give the best execution times for most applications. You may have to experiment with different options to see which combination of options generates the fastest code for your particular application.

The recommended options for generating the fastest 16-bit Intel code are:

Pentium Pro -onatz -oh -oi+ -ei -zp8 -6 -fpi87 -fp6

Pentium -onatz -oh -oi+ -ei -zp8 -5 -fpi87 -fp5

486 -onatz -oh -oi+ -ei -zp8 -4 -fpi87 -fp3

386 -onatz -oh -oi+ -ei -zp8 -3 -fpi87 -fp3

286 -onatz -oh -oi+ -ei -zp8 -2 -fpi87 -fp2

186 -onatz -oh -oi+ -ei -zp8 -1 -fpi87

8086 -onatz -oh -oi+ -ei -zp8 -0 -fpi87

The recommended options for generating the fastest 32-bit Intel code are:

Pentium Pro -onatz -oh -oi+ -ei -zp8 -6 -fp6

Pentium -onatz -oh -oi+ -ei -zp8 -5 -fp5

486 -onatz -oh -oi+ -ei -zp8 -4 -fp3

386 -onatz -oh -oi+ -ei -zp8 -3 -fp3

The "oi+" option is for C++ only. Under some circumstances, the "ob" and "ol+" optimizations may also give better performance with 32-bit Intel code.

Option "on" causes the compiler to replace floating-point divisions with multiplications by the reciprocal. This generates faster code (multiplication is faster than division), but the result may not be the same because the reciprocal may not be exactly representable.

Option "oe" causes small user written functions to be expanded in-line rather than generating a call to the function. Expanding functions in-line can further expose other optimizations that couldn't otherwise be detected if a call was generated to the function.

Option "oa" causes the compiler to relax alias checking.

Option "ot" must be specified to cause the code generator to select code sequences which are faster without any regard to the size of the code. The default is to select code sequences which strike a balance between size and speed.

Option "ox" is equivalent to "obmiller" and "s" which causes the compiler/code generator to do branch prediction ("ob"), generate 387 instructions in-line for math functions such as sin, cos, sqrt ("om"), expand intrinsic functions in-line ("oi"), perform loop optimizations ("ol"), expand small user functions in-line ("oe"), reorder instructions to avoid pipeline stalls ("or"), and to not generate any stack overflow checking ("s"). Option "or" is very important for generating fast code for the Pentium and Pentium Pro processors.

Option "oh" causes the compiler to attempt repeated optimizations (which can result in longer compiles but more optimal code).

Option "oi+" causes the C++ compiler to expand intrinsic functions in-line (just like "oi") but also sets the *inline_depth* to its maximum (255). By default, *inline_depth* is 3. The *inline_depth* can also be changed by using the C++ *inline_depth* pragma.

Option "ei" causes the compiler to allocate at least an "int" for all enumerated types.

Option "zp8" causes all data to be aligned on 8 byte boundaries. The default is "zp2" for the 16-bit compiler and "zp8" for 32-bit compiler. If, for example, "zp1" packing was specified then this would pack all data which would reduce the amount of data memory required but would require extra clock cycles to access data that is not on an appropriate boundary.

Options "0", "1", "2", "3", "4", "5" and "6" emit Intel code sequences optimized for processor-specific instruction set features and timings. For 16-bit Intel applications, the use of these options may limit the range of systems on which the application will run but there are execution performance improvements.

Options "fp2", "fp3", "fp5" and "fp6" emit Intel floating-point operations targetted at specific features of the math coprocessor in the Intel series. For 16-bit Intel applications, the use of these options may limit the range of systems on which the application will run but there are execution performance improvements.

Option "fpi87" causes in-line Intel 80x87 numeric data processor instructions to be generated into the object code for floating-point operations. Floating-point instruction emulation is not included so as to obtain the best floating-point performance in 16-bit Intel applications.

For 32-bit Intel applications, the use of the "fp5" option will give good performance on the Intel Pentium but less than optimal performance on the 386 and 486. The use of the "5" option will give good performance on the Pentium and minimal, if any, impact on the 386 and 486. Thus, the following set of options gives good overall performance for the 386, 486 and Pentium processors.

```
-onatl -oh -oi+ -ei -zp8 -5 -fp3
```

6 Release Notes for Open Watcom C/C++ 2.0

There are a number of enhancements and changes in this new version of Open Watcom C/C++. The following sections outline most of the major changes. You should consult the User's Guide for full details on these and other changes made to the compiler and related tools. You should check the next section to determine if you need to recompile your application.

6.1 Differences from Open Watcom version 1.9

Following is a list of changes made in Open Watcom C/C++ 2.0:

- OW tools ported to 64-bit host platform (Linux and Windows)
- OW installer for 64-bit host platform (Linux and Windows)

C compiler changes:

- C99 preprocessor `_Pragma()` operator was added
- `pragma "warning"` was added
- new option `-zam` was added to suppress predefined macros non-conforming ISO standard naming convention (macros without leading underscore)
- Warning W303 "Parameter '%s' has been defined, but not referenced" was enabled by default.
- Warning W308 "Unprototyped function '%s' called" was enabled by default.
- Warning W309 "Unprototyped function indirectly called" was enabled by default.
- predefined macro `NO_EXT_KEYS` was replaced by `_NO_EXT_KEYS`, old one is defined if `-zam` option is not used

C++ compiler changes:

- C99 preprocessor `_Pragma()` operator was added
- fix pragmas `"enable_message"` and `"disable_message"` to not change warning level
- new option `-zam` was added to suppress predefined macros non-conforming ISO standard naming convention (macros without leading underscore)
- predefined macro `NO_EXT_KEYS` was replaced by `_NO_EXT_KEYS`, old one is defined if `-zam` option is not used

Linker changes:

- added support for 64-bit COFF objects and 64-bit PE executable
- default stack size for Windows NT was changed to 1 MB for better compatibility with Microsoft linker

C run-time library changes:

- POSIX threads implementation was added for Linux target

Wmake utility changes:

- add interception for rm, mkdir and rmdir commands to be processed by wmake (POSIX simplified implementation)

6.2 Differences from Open Watcom version 1.8

Following is a list of changes made in Open Watcom C/C++ 1.9:

- The NULL macro is now defined as `((void *)0)` for C; for C++, NULL is still defined as 0 or 0L as previously. This change may require modifications to user code if NULL was incorrectly used as an integer constant. Note: The 16-bit windows.h header defines NULL as 0; for 16-bit Windows programming, the effective definition of NULL depends on whether windows.h or one of the standard C headers is included first.
- The C compiler has improved handling of the `_Bool` type (C99 mode specific). The `_Bool` type can now generally be used in expressions wherever an integer may be used.
- The C compiler now considers enumeration types to be compatible with their underlying integer type, as required by ISO C. Note that the choice of the underlying type varies and should not be relied upon.
- The C compiler now correctly handles initialization of static floating-point variables with 64-bit integer constants.
- The C and C++ compilers now support the `__int8`, `__int16`, and `__int32` extension keywords.
- The 16-bit C and C++ compilers now allow the `>` operator to be used in initializers of variables with static storage, as long as both operands are integer constants. This enables the use of `MK_FP` macro in initializers.
- The C++ compiler now treats warning W737, implicit conversion of pointers to integral types of same size, as an error.
- The C++ compiler now supports several new warnings: W931, warn about meaningless cv-qualifier in casts; W932, warn about meaningless cv-qualifier in function return types; and W933, warn about use of C-style casts in C++ code. These warnings must be explicitly enabled through the `-wce` option.
- The code generator now eliminates redundant epilogue code for 32-bit Intel processors when doing size optimization.

- The code generator correctly handles constant folding of signed 64-bit integer division and right shift operations.
- The code generator now supports constant folding of 64-bit integer modulo operations.
- The code generator no longer mishandles floating-point comparisons where one operand is a variable of type float and the other operand is a constant of type double or long double.
- The code generator no longer produces incorrect code when a constant expression of type float (e.g., 1.0f + 1.0f) is passed as an argument to a function which takes a float argument.
- The code generator now makes more accurate decision when choosing whether a multiplication by a constant should be replaced by a sequence of shifts and additions. Results depend on target CPU type.
- The 386 code generator now produces a CDQ instruction except when targeting a Pentium and optimizing for speed, when a MOV/SAR sequence is emitted as previously when converting a signed 32-bit integer to 64-bit.
- The code generator no longer emits redundant CS segment overrides when creating calls to symbols imported from DLLs.
- The Win32 API headers and import libraries have been updated to support the new interfaces in Windows 7.
- Support for 16-bit OLE 2.0 has been added.
- Support for RDOS targets has been added.
- Support for ZDOS targets has been added to the linker.
- The floating-point exception handler for 16-bit DOS no longer crashes if the user signal handler modified DS.
- The _floodfill() graphics library function now works correctly again.
- The library no longer leaks memory when a thread created by _beginthread() terminates.
- The %Lf format specifier (and related format specifiers) used with printf family functions now works.
- The library now contains _fseeki64 and _ftelli64 functions to handle 64-bit file offset pointer for streams.
- The library implementations of _lseeki64, _telli64, _fileleni64, _(_w)stat64, _(_w)findfirsti64, _(_w)findnexti64 on OS/2 now properly use 64-bit file sizes and offsets.
- The library implementations of puts and putws now correctly return EOF if they fail
- DOS long file name (LFN) support has been added (new doslfn?.lib model specific libraries contain DOS LFN version of appropriate modules). By default DOS LFN support is enabled. It can be switch off by setup LFN=N environment variable.

- The wasm assembler now implements support for Turbo Assembler compatible IDEAL mode. The -zcm option may be used to select this feature.
- The assembler now supports built-in @code and @data symbols.
- The assembler now supports ELSEIF directives for conditional assembly.
- The assembler no longer incorrectly assumes that absolute segments are byte aligned; they are paragraph (16 byte) aligned.
- The built-in cd command in wmake now supports quoted paths with spaces.
- The linker now supports libraries in BSD, COFF, and GNU ar format.
- The linker now allows stack sizes smaller than 512 bytes to be specified and does not silently override them. However, a new warning (W1172) will be emitted in such case.
- A FULLHEADER option specific to the DOS EXE format has been added to the linker. Please see the Linker Guide for further information.
- The library manager (wlib) now supports BSD, COFF, and GNU variants of the ar library archive format. Extended -fab, -fag, -fac options select the variant. The -fa option now selects default format depending on host platform.
- This release introduces Open Watcom's IPF compiler (wipfc) used for creating help files on OS/2.
- Documentation is now provided in Windows CHM format as an option.
- The vi editor now supports non-ASCII characters in the 0x80-0xFF range.
- The console version of the vi editor for Windows now uses standard Windows keyboard processing; AltGr and dead keys now function correctly. Additionally, Caps Lock is only effective for keys which generate characters.
- The editor now supports for syntax highlight of awk files.
- The editor now supports a Select All command.
- The windowed editor now holds the position of the replace and search/replace dialogs between invocations.
- The wcl and wcl386 utilities now work correctly on a Linux host.
- The DOS NetBIOS trap file and debug server have been modified to work when running on Windows NT (through NTVDM).
- The 16-bit Windows GUI tools no longer require CTL3DV2.DLL, but still use it if the library is present.

6.3 Differences from Open Watcom Version 1.7

Following is a list of changes made in Open Watcom C/C++ 1.8:

- The C compiler now performs stricter type checking on enumerated types. Previously, enums were treated as their underlying type (signed/unsigned char, int, etc.) for type checks.
- The C compiler now adds location information about enumerated symbols in appropriate diagnostics.
- The C compiler has been improved with respect to the handling of near/far pointers, especially when converting or comparing pointers and integers of different sizes.
- The C compiler now properly recognizes functions that do not return, such as `longjmp()`, and correctly diagnoses control flow issues such as unreachable code or missing return statements.
- The C compiler now supports a `#pragma alias`. This pragma emits alias records into the object file for processing by the linker.
- The C compiler now has larger capacity and can compile some complex source files that previously caused out of memory errors (E1064).
- The C compiler now always warns about unprototyped functions (W131). Previously, some forgivable instances of missing prototypes were undiagnosed by default, which caused users to write non-portable code usually by accident.
- Warning W139 (Divisor for modulo or division operation is zero) has been added to the C compiler. This warning is triggered whenever the right operand of an integer division or modulo operation is a constant expression with the value of zero.
- The handling of `pragma aux` is now correct in cases where code is emitted and a function body already exists for the corresponding symbolic name.
- Various fixes to the handling of the `include_alias` pragma have been made.
- The C and C++ compilers now have diagnostics for meaningless comparison of 64-bit and bit-field operands.
- The C and C++ compilers now have conversion tables from CP=1250,1252 (Latin-2,1 for Windows ANSI) to Unicode to support Windows programming.
- The C++ compiler now supports the explicit specification of function template arguments. For example: `f<T>()`.
- The C++ compiler now uses lazy instantiation for class templates and template members. Only the members actually used are instantiated.
- The C++ compiler now allows member templates to be defined outside their class.
- Numerous bugs in the C++ compiler have been fixed.
- The C++ compiler allows a class declaration with modifiers to have those modifiers propagate into the class definition.

- When -SH switch is used with the Fortran compiler, the default size of an integer constant is now `INTEGER*2`.
- The Fortran compiler no longer crashes when equivalencing common/global with automatics.
- Fortran processing for the text record EOL has been normalized. On UNIX systems write/seek uses LF and read uses LF or CRLF. On non-UNIX systems write/seek uses CRLF and read uses LF or CRLF.
- Various code generation bug fixes to both the 16-bit and 32-bit compilers have been made.
- The Win32 headers and libraries are now Open Watcom's own internal version. The MinGW headers and libraries are no longer being used. The new headers and libraries provide greater compatibility with the Microsoft SDK and better support for Microsoft Vista.
- The version of Causeway in the official distribution is now version 4.03.
- The Linux run time libraries now have `recvfrom()` and `sendto()` implementations.
- A POSIX compatible `fnmatch()` function and a corresponding `fnmatch.h` header has been added.
- The `_dos_getfileattr()` function no longer crashes in large data models on 16-bit DOS and Windows.
- The C run time library now has implementations of the following functions to improve compatibility with other compilers: `_chmod`, `_chsize`, `_creat`, `_dup2`, `_eof`, `_filelength`, `_isatty`, `_read`, `_sopen`, `_tell`, `_umask`, `_unlink`, and `_write`.
- The date and time arguments to `_dos_getftime()` and `_dos_setftime()` are now using 'unsigned int' type instead of 'unsigned short'. This change has been made to improve compatibility with other compilers.
- The segment argument used with `_dos_allocmem()`, `_dos_freemem()` and `_dos_setblock()` is now unsigned int instead of unsigned short. This change was made for compatibility with other compilers.
- The Fortran run time environment now works on Linux.
- The Linux Fortran run time libraries now have `FSYSTEM` and `FSPAWN` implementations.
- The Fortran run time now exposes the saved stack pointer for when crashes occur in the IO subsystems.
- The debugger's options context menu now has an option to allow all child nodes to be expanded recursively.
- The debugger now uses the full size of the type when displaying hexadecimal values (for example 0x03 instead of 0x3).
- The debugger now has a new menu entry to change the display format of all values in an array (all hex or all decimal).
- The debugger now supports break on write rather than just break on change. The trap must support exact breakpoints for this to happen.

- The Windows debugger now properly handles quoted program names.
- The Windows debugger now has proper color support.
- The NetWare TCP server and trap have been updated for NW6.5SP7 with the latest libcpre.obj file.
- The DOS real-mode trap file (std.trp) now correctly displays high parts of 32-bit registers on 386+ CPUs. Previously, the high parts were always displayed as zeros.
- WLIB now has a new -pa option to set up library page size automatically to optimal size.
- WLIB now handles COFF import libraries more correctly.
- WCL now properly handles the -fd and -fm options without the file name specified.
- WASM now handles EXTERNDEF directives properly; an EXTDEF record is created only if the symbol is actually referenced.
- WASM now handles the auto-dependency filename properly.
- WASM now implicitly creates the __UNIX__ macro for the BSD target as it has for LINUX and QNX.
- The internal version numbers for WASM and WMAKE are now compatible with that used by the C and C++ compilers. Specifically the macro __WASM__ has the value 1280 for WASM and the macro __VERSION__ has the value 1280 for WMAKE.
- The 32-bit DOS WD and WPROF can now be used with DOS/4G 2.x. The DOS4GOPTIONS settings are no longer exported. Users may still override the defaults by supplying their own wd.ini and wprof.ini, respectively. These files must be in the appropriate format for the DOS/4G version used. Note that this does not affect DOS/4GW users.
- WLINK now handles offsets and groups larger than 64 KB for 32 bit code and 16 bit targets.
- WLINK now ignores fixup displacement when the target relocation is absolute. This is required for compatibility with object files generated by MASM 5.1.
- WLINK now properly handles the alignment of the last segment in a group if the last segment fragment is blank.
- WLINK can now use the WLINK_LNK environment variable to override the default directive file name (wlink.lnk). If the specified file isn't found then default file is used as usual.
- WLINK now properly emits segments overlapped by groups to output file.
- WLINK now properly handles imported symbols that are locally defined with the dllimport specifier in PE formatted files.
- WLINK DLL was renamed from wlink.dll to wlinkd.dll. It requires to correct wmake directive "!loaddll" to use this new name if it is used in makefile.
- WRC on Far-Eastern NT-based systems now honors the DBCS encoding specified on the command line.

- The text editor now supports syntax highlighting for resource files.
- The Fgrep dialog box in graphical editor now has a browse button to display the standard browse for folder dialog box on versions of Windows that support it.
- The ide2make utility has been added to the Open Watcom distribution. This utility converts IDE project files to make files.
- The exe2bin utility now performs properly if relocation items are unsorted.
- The installer now generates a batch file that automatically sets the environment variables used by the Open Watcom tools.

6.4 Changes in 1.8 that may Require Recompilation

_dos_gettime and _dos_settime functions 32-bit users of `_dos_gettime()` and `_dos_settime()` need to recompile. Consider using `stat()/utime()` instead of the non-portable functions.

_dos_allocmem and _dos_freemem functions 32-bit DOS users of `_dos_allocmem()` and `_dos_freemem()` need to recompile. Consider not using these functions in the first place.

6.5 Differences from Open Watcom Version 1.6

Following is a list of changes made in Open Watcom C/C++ 1.7:

- Functions using the `__cdecl` calling convention now return floating-point values in FPU registers in 32-bit code. This is the correct behavior compatible with other compilers.
- The C and C++ compilers now support an `include_alias` pragma which can be used, among other things, to map long include filenames onto 8.3 names. In addition, the compilers now support a `-na` switch to disable the automatic inclusion of `_ialias.h` (which contains new `include_alias` pragmas for certain standard headers).
- The C compiler now emits warnings on operations which mix multi-level pointer types such as `void**` and `int**`. Previously a warning was only emitted in ANSI mode. Note that while `void**` is not assignment compatible with `int**`, `void*` is. The new stricter behavior is in line with most other C compilers and helps users write cleaner and more portable code.
- The C compiler now warns (W400) when an expression of plain char type is used as array index. Because the sign of plain char changes depending on compiler choice and settings, using plain char as array index is inherently non-portable and likely to cause problems.
- The C compiler now folds constant expression before checking for range overflows etc. This enables the compiler to diagnose constant out of range issues in more cases.
- The C compiler now allows benign variable redefinitions where `__near` or `__far` modifiers aren't always explicitly specified but match when current data model is taken into account (eg. `'extern int __near x;'` and `'extern int x;'` in small data models). The new behavior is compatible with the C++ compiler as well as with other vendors' compilers.

- The default stack size for 32-bit DOS extended executables has been increased from 4K to 64K. Note that the linker can still set greater or smaller stack size.
- The code generator now produces better 16-bit code for 64-bit math operations
- The code generator no longer incorrectly optimizes out conditionals. The problem only occurred in very rare situations and probably only when doubles were being compared.
- A long-standing problem with use of certain 64-bit constants as results of ternary operators has been fixed.
- The C++ compilers now has a workaround for bugzilla bug #63 (http://bugzilla.openwatcom.org/show_bug.cgi?id=63). The compiler now generates an error message referring to the bug entry rather than crashing.
- The LIBC and CLIB thin network libraries have been added to the distribution as experimental.
- 32-bit DOS executables now correctly pass environment to child processes started through the spawn family of functions. Previous incorrect behavior was introduced in version 1.6.
- In the NetWare libraries, `__get_stdout` with `__get_std_stream` was causing an infinite recursion. This has been fixed.
- Fixed 8087 emulator/mathlib 80-bit real multiplication if one operator is zero and second is a power of two.
- The `_outtext()` function in `graph.lib` no longer fails on 32-bit targets.
- The Win32 `stat()` function now returns correct results when it tries to access a file with given name and the directory containing that file also contained a subdirectory with the same name.
- The debugger now supports a "No Source" toggle (right mouse click menu) in the assembly code window to switch on/off associated source code.
- The debugger no longer crashes when tracing F77 programs that use variable-size arrays. This only applies to DWARF debugging information (which is used by default).
- The debugger now correctly displays multi-dimensional Fortran arrays when DWARF debug information format is used (which is used by default).
- The debugger can now display [partial] strings in Fortran code when the string length exceeds the debugger's internal limit. The internal limit has also been increased from 512 to 1024 characters.
- The resource compiler now properly copies non-resident name table when processing LX executables.
- The console version of vi for Win32 no longer quits after Ctrl+Left is pressed followed by any other key.
- WCL now correctly takes options from the environment when there is more than one file to compile.
- The linker now supports a MIXED1632 option to allow mixing of 16- and 32-bit logical segments into a single physical segment in OS/2 LX/LE executables.

- The linker now supports a NOSTUB option for Windows and OS/2 executable formats. This option causes no DOS stub executable to be written to the output image.
- The installer now allows you to disable creating program groups or modifying the startup environment using the /np and /ns switches.

6.6 Changes in 1.7 that may Require Recompilation

`__cdecl` Calling Convention Functions in 32-bit code that return floating-point values now do so using the FPU registers. Such functions will need to be recompiled if they are to be linked with object code produced by Open Watcom C/C++ 1.7.

6.7 Differences from Open Watcom Version 1.5

Following is a list of changes made in Open Watcom C/C++ 1.6:

- The C compiler has been modified to use the underlying bit-field type and not signed/unsigned int as the type of operand which is a bit-field. This is consistent with the C++ compiler and fixes some problems when bit-fields larger than int are used.
- Processing of `#pragma aux` has been corrected in the C compiler. This fixes problems when using the `mmintrin.h` header, among others.
- The C compiler now accepts `__declspec` modifiers specifying calling conventions applied to variables, not just functions. The new behavior is consistent with the C++ compiler, and also with the fact that ordinary calling convention type modifiers can be used with variables.
- The C and C++ compilers have been fixed to properly declare variable names based on calling convention specifiers. This fixes problems with building code using IBM SOM. Note that the current behavior is the same as in Open Watcom 1.3 and earlier.
- The C compiler's preprocessor has been modified to allow use of macros with large number of arguments (255 or more).
- The C compiler no longer generates internal errors when options `-ri` and `-oe` are specified at the same time.
- The C++ compiler has been fixed to inline intrinsic functions.
- The 386 compilers have been changed to default to tuning code for P6 architecture instead of Pentium. Optimizing for P6 typically results in slightly more compact and faster code.
- The 386 C compiler has been fixed to properly convert between flat and `__far16` pointers, especially pointers to functions. Its behavior should now be compatible with the C++ compiler. The problem was most likely affecting OS/2 users who wrote mixed 16-bit and 32-bit code.
- The C compiler has been changed to allow redeclaration of functions in rare cases where initial declaration did not specify a calling convention and the subsequent declaration specified a calling convention which matched the default.

- A new -zwf switch has been added to the C and C++ compilers. This switch is off by default and enables generation of FWAIT instructions on 386 and later CPUs. It is only needed in unusual situations.
- The C compiler now correctly converts 64-bit integer constants to floating-point constants.
- The code generator no longer merges memory accesses when volatile variables are involved.
- The code generator now correctly const folds 64-bit right shifts.
- The code generator now properly converts between far pointers and 64-bit integers. Attempts to convert a 48-bit far pointer to 64-bit integer no longer cause a crash.
- The code generator has been modified to slightly decrease code size when optimizing for size (-os).
- The non-standard alloca.h header has been added for compatibility with other compilers.
- The strftime() library function has been extended to support date formats introduced in C99.
- The file pointer type used with lseek() and tell() has been changed to off_t (from long) for compatibility with POSIX.
- The 386 versions of _clear87() and _status87() functions have been modified to use the no-wait form of FPU control instructions. This allows these functions to be used in exception handlers when there are pending unmasked floating-point exceptions.
- The 16-bit 8087 emulator has been fixed to correctly evaluate multiplies as infinity instead of zero in rare overflow situations.
- The resource compiler (wrc) has been fixed to store long integer constants as 32-bit quantities in RCDATA or user data resource statements. This behavior applies to Win16, Win32, and OS/2 targets. Integers without the 'L' suffix are stored as 16-bit and potentially truncated.
- The OS/2 specific part of the resource compiler has been corrected to process RCDATA statements properly.
- The assembler (wasm) now supports external absolute symbols. The SIZE, SIZEOF, LENGTH, and LENGTHOF operators have been corrected for structures.
- Classification of privileged instructions in the assembler has been updated to match MASM.
- The assembler now evaluates expressions in return instructions correctly. Previously, code such as 'ret 28+4' would be sometimes erroneously assembled as 'ret 28' instead of 'ret 32'.
- The linker has been changed to only recognize segments of class 'STACK' as stack segment. Previously, any segment with class name ending with 'STACK' (eg. 'FSTACK') was recognized.
- Several minor problems related to creating DOS executables have been fixed in the linker.
- The RUNTIME linker directive has been extended to allow ELF ABI type and version specification. This functionality is similar to the brandelf utility. See the Linker Guide for details.

- The wmake utility has been modified such that in native wmake mode, a symbolic target with no command list is always considered to have had its command list executed. That will cause any targets that are dependent on this symbolic target to be considered out of date.
- The Win32 trap file is now able to determine the full pathname of debuggee's loaded DLLs. This may ease debugging in some cases as the debugger will be more likely to find debugging information for DLLs.
- The Win16 debugger trap file (std.dll) has been modified to allow 16-bit wdw to run on Windows NT platforms without reporting a spurious error message on exit.
- Numerous problems with the Win386 extender support have been fixed so that Win386 now works again.
- The dmpobj utility has been enhanced to support additional OMF records, and new command line options have been added.

6.8 Differences from Open Watcom Version 1.4

Following is a list of changes made in Open Watcom C/C++ 1.5:

- Support for ISO/IEC TR 24731, "Extensions to the C Library, Part I: Bounds-checking interfaces" has been added to the C runtime library. The C compiler now predefines the macro `__STDC_LIB_EXT1__` (which evaluates to 200509L) to indicate this support. This set of functions is also known as the Safer C Library. Please see the C Library Reference for detailed documentation of these functions.
- In C99 mode, the C compiler now supports C99 style declarations intermixed with statements within a block, as well as declarations in the opening clause of a for loop.
- The C compiler now predefines additional macros required by the C standards. These include `__STDC_HOSTED__` (evaluates to 1) to indicate a hosted implementation and `__STDC_VERSION__` (either 199409L or 199901L) to indicate C94 or C99 support depending on compilation switches.
- A `__restrict` keyword has been added to the C compiler. It is functionally equivalent to the C99 'restrict' keyword but is always visible, even in non-C99 mode.
- In C99 mode, the C compiler no longer requires explicit return statement in a main function returning an integer. In accordance with C99 standard, 'return 0;' is implied if end of function is reached and no return statement was encountered.
- The C compiler has been fixed so that it no longer loses track of pointer base when using `__based(__self)` pointers. The C compiler has also been fixed to properly handle dereferencing of based arrays.
- The 16-bit C++ compiler now correctly casts pointers to `__segment` type without crashing. Proper semantics of the cast were implemented.
- The 16-bit C compiler has also been modified to handle casts to `__segment` type correctly, ie. return segment portion of a pointer.

- The C compiler has been fixed to properly support named based segments (ie. `__based( __segname( <name> ) )` pointers) that reference the default `_DATA` and `_CONST` segments.
- The compilers and library headers have been modified to properly support use of standard libraries when one of the `-ec` switches is used to change default calling convention.
- The C compiler has been modified such that segments created through `#pragma data_seg` are no longer part of `DGROUP`. Typically, the pragma is used only when data segments need to be located in a physically separate segment in the output module.
- New warning W137, "Extern function 'fn' redeclared as static", has been added to the C compiler. Existing error E1072, "Storage class disagrees with previous definition of 'symbol'" has been extended to cover redefinitions from 'extern' to 'static' and not only from 'static' to 'extern'. Changing the linkage of a symbol invokes undefined behavior according to ISO C.
- New warning W138, "No newline at end of file", has been added to the C compiler. It is emitted if no line terminator character was found before the end of a source file. Such files do not conform to ISO C. The missing newline character will be automatically inserted; this matches the C++ compiler behavior. Note that missing newlines could previously lead to spurious "#endif matches #if in different source file" warnings.
- The C compiler has been modified to allow the `__export` or `__declspec(dllexport)` modifier on a declaration when earlier declaration exists with no modifier. The updated behavior is compatible with the C++ compiler as well as some compilers from other vendors.
- In ISO/ANSI mode (`-za`), the compiler now always warns if it encounters a call to unprototyped function. In extensions mode (default, `-ze`), this warning (W131) is suppressed if a matching prototype is found later in the source file. Note that the behavior in extensions mode is unchanged from earlier versions.
- The C compiler now eliminates static functions that are always inlined from the object file (the functions are of course still emitted if their address is taken).
- The C compiler has been fixed to properly evaluate boolean expressions (especially the ternary operator) where the condition is a 64-bit integer constant. Previously, the high 32 bits were in some cases ignored, which could lead to erroneous results.
- The C compiler has been modified to properly cast floating-point constants to the specified type. Notably `FLT_MIN` stored or passed as double is now handled correctly (without spurious precision).
- Handling of empty macro arguments has been corrected in the C compiler's preprocessor. Previously, empty macro arguments could result in invalid tokens in certain cases.
- The peephole optimizer is now run again after register allocation. This allows the code generator to take advantage of some optimization opportunities that were previously missed.
- The code generator has been modified to copy `DS` into `ES` in `__interrupt` routine prolog (right after `DS` was loaded) if the `ES` register is considered unalterable, ie. in flat model. This may avoid crashes if `ES` is dereferenced explicitly or implicitly in interrupt handlers, for instance when calling `memcpy()`.
- The linker and other tools have been fixed to correctly classify code segments. Previously, code segments could be misclassified as data, which led to incorrect disassembly and generation of debugging information.

- A performance problem related to emitting debugging information for structures or unions with many members has been corrected in the code generator.
- The POSIX-defined header `libgen.h` has been implemented. This includes two functions, `basename()` and `dirname()`.
- The functions `btowc()`, `fwide()`, `mbsinit()`, `wctrans()`, and `towctrans()` have been added to the C runtime library. These functions are all related to wide-character and multi-byte support, and were first defined by the ISO C Normative Amendment 1.
- C99 functions `llabs()`, `lldiv()`, and `_Exit()` have been added to the C runtime library. Note that the latter is equivalent to `_exit()`, defined by POSIX.
- Support for C99 floating-point classification macros has been implemented. This includes `fpclassify`, `isfinite`, `isinf`, `isnan`, `isnormal`, and `signbit`.
- Modifiers `'hh'`, `'j'`, `'z'`, and `'t'` defined by C99 for the `printf` and `scanf` family of functions have been implemented in the C runtime library. Please see the C Library Reference for details.
- The `'F'` modifier for `printf` and `scanf` families of functions conflicts with `'F'` format specifier defined by ISO C for floating-point conversions. It has been replaced by a `'W'` modifier which is now used to denote a far pointer. The `'F'` modifier is still recognized in DOS builds of the runtime library (which therefore cannot handle the `'F'` format specifier as defined by ISO C), but is no longer documented and will be removed in a future release.
- Several very obscure bugs have been fixed in the `printf` and `scanf` family of functions. These problems were discovered thanks to a more stringent testing procedure and had never been reported by users.
- The `strtod` function has been enhanced to follow C99 specification. It will now parse infinity and NaN, as well as hexadecimal floating-point numbers. See the C Library Reference for details.
- The math library has been fixed to perform binary to decimal floating-point conversions with greater precision. This fixes a problem where in some cases a conversion from binary to decimal and back was losing precision or producing erroneous results.
- The graphics library has been fixed to correctly work with VESA modes where the number of bytes per line does not directly correspond to width of the mode in pixels.
- The `owcc` utility has been much improved and documented; this tool is a POSIX style compiler driver, designed to provide certain level of command line compatibility with `gcc` and ease porting.
- The `NOEXTension` linker option has been documented; this option instructs the linker not to add any extension (`.exe`, `.dll`, etc.) to the executable name. Any argument to the `NAME` directive will be used verbatim. This option had been supported by earlier versions of the linker but not documented.
- The `'include'` preprocessor directive not prefixed by an exclamation mark is now recognized in `wmake -ms` mode for compatibility with Microsoft and IBM `NMAKE`.
- The `wmake` utility has been enhanced to evaluate `NMAKE` style `'[cmd]'` expressions (ie. shell commands) in preprocessor `!if` directives. This functionality is supported in both `wmake` and `-ms` mode.

- Several problems related to the Watcom debugging information format have been fixed in the linker and debugger. Note that the Watcom format is considered obsolete and its use is not recommended.
- A random but very rare startup failure of Windows based GUI tools (notably wdw) has been fixed.

6.9 Differences from Open Watcom Version 1.3

Following is a list of changes made in Open Watcom C/C++ 1.4:

- Support for C99 designated initializers has been added to the C compiler, for example "struct {int a, b;} c = {.a=0, .b=1};". This is also supported for arrays, for example "int a[4] = {[0]=5, [3]=2};".
- Handling of enumerations has been fixed in the C compiler. In certain cases, the compiler chose the wrong type for operations on objects of enumerated types. Enumerated constants up to 64 bits wide are now also allowed (including in 16-bit compilers).
- The C compiler will now warn if the right hand operand of a bitwise shift expression is a constant that is either negative or greater than or equal to the bit width of the promoted left operand. The result of such operation is not defined by ISO C. The warnings are 'W134: Shift amount negative' and 'W135: Shift amount too large'.
- The C compiler now warns in cases where an unsigned type is compared for <= 0. This is equivalent to 'unsigned == 0' and often indicates that a signed comparison was intended.
- New `__watcall` keyword has been added to the C and C++ compilers to designate the default Watcom calling convention.
- New `-ec?` switches have been added to set the default calling convention.
- The 16-bit C compiler now defines `_M_I86` macro for consistency with the C++ compiler. The new macro should be used in preference to the existing `M_I86`.
- A number of new keywords have been added to the C compiler; these were previously defined as macros: `_Cdecl`, `_Export`, `_Far16`, `_Fastcall`, `_Pascal`, `__syscall`, `_System`, `__try`, `__except`, `__finally`, `__leave`.
- Analogous change has been made to the C++ compiler. The new keywords (and removed predefined macros) are: `_Cdecl`, `_Export`, `_Far16`, `_Fastcall`, `__inline`, `_Pascal`, `__syscall`, `_System`.
- The C++ compiler now handles the "new" template specialization syntax, and partial specialization is partially supported (no pun intended).
- The C++ compiler now correctly handles the situation where control reaches the end of `main()` function without encountering a return statement. In that case, the effect is that of executing "return 0;".
- The C++ compiler now properly allows return statements with a void expression in functions that return void.
- 386 C and C++ compilers now support the Microsoft fastcall calling convention, and recognize the `__fastcall` keyword.

- The C compiler now recognizes `#pragma data_seg` and `code_seg` forms that specify segment and class names without enclosing them in parentheses. The new behavior is consistent with other compilers.
- New `-fti` switch has been added to the C compiler to track `#include` file opens. This helps diagnose include file problems in complex projects.
- The code generator no longer emits debug information for unreferenced typedefs when `-d1+` or `-d2` switch is used. This produces slightly to significantly smaller debug information. Note that behavior of `-d3` is unchanged.
- The 386 code generator will no longer select the `'and'` instruction to perform zero extension when optimizing for time on the 686 architecture (`-6r` or `-6s` switch). The `'movzx'` instruction will always be used instead, because it avoids partial register stalls and in certain cases significantly improves performance on P6 and newer class CPUs.
- Support for long long based bitfields has been improved in the code generator.
- The code generator now properly diagnoses attempts to emit symbol names that overflow the OMF limit (255 bytes).
- Several problems related to loss of segment information in 386 non-flat models have been fixed in the code generator.
- Command line processing has been changed in the Compile and Link utility (`wcl`). Forward slashes now may be used as path separators in file arguments, such that `"foo/bar"` is now interpreted as `"foo\bar.c"`. Note that this does not affect options delimited with forward slashes.
- Support for raw binary and Intel Hex output has been added to the linker, along with support for 24-bit segmented addressing architectures (HSHIFT option) and arbitrary class/segment reordering (OUTPUT and ORDER directives). See Linker Guide for details.
- Support for overlays (16-bit DOS) has been reinstated in the linker. A related FARCALLS/NOFARCALLS option has been added to the linker. See Linker Guide for details.
- The linker now correctly processes relocations to symbols in absolute segments.
- The linker now checks for bitness conflicts (16 vs. 32-bit) when adding segments to a group.
- The minimum accepted value for linker OBJALIGN option has been changed to 16 bytes (previously 512).
- The linker no longer creates a separate `.bss` section when linking PE modules. Instead, uninitialized data is added at the end of the data section. This creates slightly smaller executables and reduces memory usage.
- The `st_name` member was removed from struct `stat` and related structures. This was done for consistency across platforms (UNIX has no such field), because the `st_name` field was almost entirely useless (being limited to 13 characters), and for compatibility with Microsoft compilers; the latter because struct `_wstat` and struct `_wstati64` are now obsolete and struct `_stat/_stati64` can be used for wide character stat functions. NB: This change requires recompilation. New object files will not work with old libraries and vice versa.

- The `signal()` function no longer modifies floating-point control word on Win32 and 32-bit OS/2. Also, default value for `SIGFPE` has been changed to `SIG_DFL` from `SIG_IGN` on all platforms.
- The `e/E` format of `printf()` family of functions has been changed to format the exponent with minimum of two digits (instead of three). This behavior is dictated by C99 and consistent with most other compilers.
- The floating-point to string conversion routines now format values with greater precision. This means that floating-point values printed by C and C++ programs may be slightly different from earlier versions of the runtime libraries (but more accurate).
- The `sleep()` function is now declared in `unistd.h` and its return type has been changed to unsigned int, for compatibility with POSIX.
- The `clock()` function now uses millisecond counters (where available) on DOS and Windows, and is no longer susceptible to problems related to TZ changes.
- The DOS runtime has been tuned to produce smaller executables.
- C99 functions `wmemchr()`, `wmemcmp()`, `wmemcpy()`, `wmemmove()`, and `wmemset()` have been added to the C runtime library.
- A POSIX compatible `getopt()` function has been added to the C runtime library.
- A POSIX compatible `mkstemp()` function has been added to the C runtime library.
- BSD compatible safe string copy and concatenation functions, `strncpy()` and `strncat()`, have been added. Use of these functions is highly recommended over `strncpy()` and `strncat()`, because they are safer and much easier to use.
- New `strings.h` header has been added for POSIX compatibility, although legacy functions `index()` and `rindex()` are not supported. Functions `strcasestr()` and `strncasestr()` are also declared in `string.h` for compatibility with other compilers.
- The C runtime library no longer returns `ESPIPE` when calling `write()` on a pipe or device that was opened with `O_APPEND` flag. The old behavior was not POSIX conforming.
- Handling of pathnames that include spaces has been improved in the make utility (`wmake`).
- The disassembler (`wdis`) now handles big endian object files on little endian host platforms, and vice versa.
- Support for MIPS R4000 and SPARC V8 instruction sets has been added to the disassembler.
- New `-zz` and `-zzo` option have been added to the assembler (`wasm`) for backwards compatibility. See Tools User's Guide for details.
- Default behavior of inline assembler has changed. The CPU optimization level (`-4`, `-5`, `-6`) now implies the available instruction set: `-5` implies MMX and 3DNow!, `-6` also implies SSE/SSE2/SSE3. Also note that any CPU setting override now reverts to default at the end of each inline assembly block.
- 16-bit DOS version of the assembler (`wasmr`) has been added. This version runs on 8086 and above and requires less memory than the protected mode version.

- The debugger has been changed to look for support files in directories relative to the debugger executable's location. This allows the debugger to be used when no debugger specific environment variables have been set.
- A problem with stepping into code (F8) right after debuggee was loaded has been fixed in the debugger.
- The debugger now looks for debug information in a .sym file when the /Download option was specified. Previously it erroneously only looked at the executable if the download option was used.
- Support for Microsoft/IBM .sym files generated by the MAPSYM utility has been added to the debugger and profiler. This is helpful especially with symbol files provided by IBM for OS/2 system DLLs; disassembly now shows for instance "call DOS32EXIT" instead of "call 01C74634".
- The CauseWay trap file no longer incorrectly maps symbol addresses in 'large' executables (code segment > 64K).
- Interoperability with GNU tools has been improved. The debugger (wd/wdw) should now be able to debug GNU-produced executables (with DWARF 2 debug information) and vice versa.
- New -zld option has been added to the library manager (wlib) to strip autodependency information from OMF objects.
- New exe2bin utility has been added. See Tools User's Guide for details.
- Basic support for compiling OS/2 resource scripts and binding resources into OS/2 executables (both NE and LX formats) has been added to the resource compiler (wrc).
- The include search order in the resource compiler has been changed to be more consistent with the C/C++ compilers, as well as with IBM's and Microsoft's resource compilers. System include files (enclosed in angle brackets) are no longer searched in current directory or in the directory of the file containing the #include directive.
- The Windows resource compiler has been made more compatible with scripts designed for Microsoft's RC in the way it treats string literals.
- The MS LINK compatibility wrapper now supports a /RELEASE switch.
- Syntax highlighting support for makefiles has been added to the editor. The default syntax highlighting scheme has also been made more colorful.
- The editor and Windows GUI tools now store configuration files in more appropriate locations (notably on multi-user machines).
- Several new DOS extender related targets have been added to the IDE. Better support for remote debugging has also been added to the IDE.
- The CauseWay DOS extender now supports SSE instructions on plain DOS.
- Several simple OS/2 SOM programming examples have been added.

6.10 Changes in 1.4 that may Require Recompilation

stat() The *stat* function now uses a slightly different *struct stat* argument. Source code that uses the *stat* function or references *struct stat* must be recompiled before linking the application with new libraries.

6.11 Differences from Open Watcom Version 1.2

Following is a list of changes made in Open Watcom C/C++ 1.3:

- The C++ compiler now restricts the scope of variables declared in a for loop to the scope of that loop in accordance with ISO C++, not extending the scope beyond the loop (ARM compliant behavior). Code relying on the pre-standard behavior must either be changed or compiled with new -zf switch which reverts to old scoping rules.
- Support for default template arguments has been added to the C++ compiler.
- Support for alternative tokens (and, xor etc.) has been added to the C++ compiler. It is enabled by default, can be turned off with the new -zat switch.
- The C runtime library has been made significantly more C99 compliant. A number of new headers have been added (inttypes.h, stdbool.h, stdint.h, wctype.h) and corresponding new functions implemented. Wide character classification functions were moved out of ctype.h into wctype.h. C99 va_copy macro was added to stdarg.h.
- Added 'cname' style C++ headers.
- Support for SSE, SSE2, SSE3 and 3DNow! instruction sets has been added. Affected tools are the assembler (wasm), as well as all x86 compilers, disassembler and debugger. The debugger now also supports MMX registers formatted as floats (for 3DNow!) as well as a new XMM register window for SSE.
- Inline assembler directives .MMX, .K3D, .XMM, .XMM2 and .XMM3 are now supported in the _asm as well as #pragma aux style inline assembler interface. Note: .MMX directive is now required (in addition to .586) to use MMX instructions.
- C compiler performance has been significantly improved (up to 5-10 times speedup) when compiling large and complex source files.
- All x86 compilers now have the ability to perform no truncation when converting floating point values to integers. Additionally, 32-bit x86 compilers have the option to inline the rounding code instead of calling __CHP.
- The C lexical scanner no longer evaluates constants with (U)LL suffix that fit into 32 bits as zero (1ULL was wrong, LONGLONG_MAX was correct).
- C and C++ x86 inline assembler has been fixed to properly process hexadecimal constants postfixed with 'h'.
- The C compiler now supports the C99 'inline' keyword, in addition to previously supported '_inline' and '__inline' keywords.

- The C compiler now treats a sequence of adjacent character strings as wide if any of the components are wide (required by C99), instead of depending on the type of the last component. For example, `L"foo " "bar"` is now interpreted as `L"foo bar"`, instead of `"foo bar"`.
- The internal C compiler limit on complex expressions has been increased and if it is still insufficient, the compiler now reports an error instead of crashing.
- The C compiler now issues a warning on the default warning level if a function with no prototype is referenced. This was previously warning W301 (level 3), now it is warning W131 (level 1).
- Warning "W132: No storage class or type specified" has been added to the C compiler. This warning is issued if a variable is declared without specifying either storage class or type. This is not allowed in C89.
- Warning "W304: Return type 'int' assumed for function 'foo'" has been added. This warning is issued if a function is declared without specifying return type. This is allowed in C89 but not in C99.
- Warning "W305: Type 'int' assumed in declaration of 'foo'" has been added to the C compiler. This warning is issued if a variable is declared without specifying its type. This is allowed in C89 but not in C99. Note that if warning W132 is issued, W305 applies as well.
- The C compiler now properly warns if a function with implied 'int' return type fails to return a value. This potential error was previously undetected.
- C++ compiler diagnostic messages have been made more consistent and slightly more detailed.
- Linker for Win32 targets can now create file checksums. These are primarily used for DLLs and device drivers, but can be applied to all Win32 PE images if required.
- Linker for Win32 targets can now set operating system version requirements into the PE/COFF optional header (Microsoft extended header).
- Linker for Win32 targets can now set the linker version number into the PE optional header (Microsoft extended header).
- The linker will now eliminate zero-sized segments from NE format (16-bit OS/2 and Windows) executables. This fixes a problem where Windows 3.x would refuse to load an executable with zero sized segment. This could happen especially with C++ programs where some segments may have ended up empty after eliminating unused functions.
- The linker now (again) produces correct Watcom style debugging information. This was a regression introduced in previous version.
- Command line parsing for `wccxxx`, `wppxxx` and `cl` has been changed such that a double backslash inside a quoted string is collapsed to a single backslash, and hence `"foo\\"` now translates to `'foo\'` and not `'foo\'`.
- The IDE and other graphical tools no longer leak system resources (a bug introduced in version 1.2).
- The Image Editor limit on bitmap size has been changed from 512x512 pixels to 2048x2048 pixels.
- The source browser now correctly decodes array information; Version 11.0c of Watcom C/C++ started emitting array browse information in a new format and the browser hadn't been updated accordingly.

- The NT debugger trap file has been changed so an exception generated during a step operation is handled correctly. Previously, the single step flag was not being cleared and when the exception was being offered to the debuggee's own exception handlers, a single step exception occurred in NT's exception handler rather than the exception being passed back to our handler.
- The OS/2 debuggers now dynamically allocate buffer for the command line, preventing crashes when the command line was over approx. 260 bytes long.
- The NetWare 5 debugger NLM has been changed to use kernel primitives. Previous version were using legacy semaphores.
- The make program (wmake) has been sped up very slightly. Also the 'echo' command is now internal and no longer spawns the system command interpreter.
- The precision of DBL_MAX, DBL_MIN and DBL_EPSILON has been increased; the non-standard variants prefixed with an underscore have been removed.
- The C99 functions atoll(), ltoa(), ulltoa(), strtoll(), strtoull() and corresponding wide character functions have been added to the C runtime library.
- The _beginthread() function now consistently returns -1 in case of error on all platforms.
- The stdaux and stdprn streams are now only defined on DOS based platforms, ie. DOS, Win16 and Win386. No other platforms support stdaux or stdprn.
- The assert() macro now prints function name in addition to source file and line number, in accordance with C99.
- The _heapchk() function will now always perform a consistency check on the heap, where it would previously only check consistency if there had been allocations/frees since last call to _heapchk(). As a consequence, _heapchk() previously did not detect certain instances of heap corruption.
- [OS/2 32-bit] The default __disallow_single_dgroup() implementation no longer statically links against PMWIN.DLL. This allows DLLs to load on systems where PMWIN.DLL isn't present.
- [OS/2 32-bit] Re-implemented clock(). The new implementation uses the OS millisecond counter and is hence not susceptible to TZ changes. It is also smaller, faster and more accurate, although it may wrap around earlier than the original implementation.
- The disassembler (wdis) now correctly processes x86 'push 8-bit immediate' instructions.
- The disassembler now correctly processes absolute memory references. All memory references without fixup are now disassembled as ds:[...] or sreg:[...].
- Several DirectX Win32 programming samples have been added. Note that a separate DirectX SDK (available from Microsoft) is required to build these sample programs.

6.12 Differences from Open Watcom Version 1.1

Following is a list of changes made in Open Watcom C/C++ 1.2:

- Handling of default libraries for Win32 targets has been changed. Previously default library records were included in the C runtime library, now they are specified part of the wlink Win32 target definitions in wlsystem.lnk. The list of libraries has changed from previous version as well and now reflects the Microsoft compiler defaults; existing IDE projects and makefiles may need to have additional libraries specified if those are no longer part of the default list.
- The C compiler now performs stricter checking on function prototypes and pointer operations. This may lead to previously undiagnosed warnings/errors appearing when compiling incorrect or ambiguous code.
- The C compiler diagnostic messages have been improved to print more information, making it easier to isolate the problem.
- A new warning (W130) has been added to the C compiler to diagnose possible precision loss on assignment operations. This warning is never on by default and must be enabled through `'#pragma enable_message(130)'` or `'-wce=130'` switch.
- Support for C99 style variable argument macros (and `__VA_ARGS__`) has been added to the C and C++ compilers.
- Added support for the `__func__` symbol (name of the current function) which is equivalent to the already existing `__FUNCTION__` symbol.
- Better C99 style support for "long long" type is now available in the C and C++ compilers. LL, ULL and LLU suffixes are recognized for constants. "long long int" is now also recognized.
- Added C99 style `*LLONG_MIN/MAX` defines to limits.h.
- The C++ compiler has been fixed to properly accept source files where a template was the last item in a name space or an external linkage.
- Several new `-adx` options have been added to the C and C++ compilers to support automatic generation of 'make' style dependency files.
- The C compiler has been fixed to correctly diagnose illegal union assignments.
- The C compiler now issues warnings on operations involving pointers to different but compatible unions.
- The C and C++ compilers now ensure word alignment of wide character string literals to satisfy Win32 API restrictions.
- The `__UNIX__` macro is now supported in C and C++ compilers, wmake and wasm. It is currently defined for QNX and Linux targets.
- Default windowing support has been re-enabled for Win16 and Win386 runtime libraries.
- Since default windowing is no longer supported on most platforms, the Programmer's Guide and IDE tutorial have been updated to reflect that fact.

- The Win32 GUI tools now support the Windows XP look and feel.
- AutoCAD and MFC targets have been removed from the IDE, the -bw switch (default windowing) is no longer available in the IDE for OS/2 and Win32 targets.
- Manual for the CauseWay DOS extender has been added.
- The dmpobj tool has been added. This utility dumps the contents of OMF object files and can be useful to developers.
- Several system definitions have been added to wlink: os2_pm (16-bit OS/2 Presentation Manager executable), os2_dll (16-bit OS/2 DLL) and os2v2_dll (32-bit OS/2 DLL).
- The linker has been fixed to read "AR" style archives produced by third party tools.
- The linker has been fixed to prevent crashes when linking with COFF files providing uninitialized COMDAT entries
- Several linker crashes related to ELF object files and executables have been resolved.
- Updated wlink to call wlib with the -c (case sensitive) option when creating import libraries. This fixes problems with DLLs that export symbols differing only in case.
- The C runtime library has been optimized to produce smaller executables.
- The printf() function now supports the "ll" format specifier for "long long" integers.
- The printf() function has been enhanced to support %b format specifier for bitfields.
- Execution order of C library termination routines is now better defined to prevent instances where temporary files created through mktemp() could be left behind.
- [OS/2 32-bit] To prevent crashes, termination code is not run if second instance of a DLL failed to load due to single DGROUP.
- [OS/2 32-bit] The __grow_handles() function was incorrectly adding n requested handles to existing limit instead of setting the limit to n.
- [OS/2 32-bit] Fixed a problem with _STACKLOW in multithreaded programs and DLLs. This prevents crashes where Fortran DLLs would run out of stack.
- [OS/2 16-bit] Fixed default math exception handler which wasn't popping the FP status word off the stack and would therefore crash on exit.
- The Win32 Image Editor has been enhanced with drag-and-drop support.
- The IDE has been fixed to properly handle mixed case filenames.
- The Microsoft compatibility tools (NMAKE, CL) have been fixed to better handle command line arguments.
- The Dialog Editor (wde) has been fixed to prevent occasional DDE related crashes when run from inside the Resource Editor (wre).

- The 'Change font' option no longer crashes the GUI debugger (wdw).
- On OS/2, wdw now intercepts the F10 key instead of passing it on to the system.
- The code generator now deletes object files if it was interrupted. Previously zero-length invalid object files could be left behind, interfering with make operation.
- The wasm assembler has been enhanced to generate file dependency information usable by wmake.
- Numerous minor fixes have been made to wasm.
- Compatibility with MASM 6 has been improved with wasm.
- Support for sysenter and sysexit instructions has been added to wasm and wdis.
- Disassembly of xchg and bound instructions has been fixed in wdis (corrected order of operands).
- Several previously undocumented wmake directives have been documented.
- A -sn ('noisy') option has been added to wmake to print all commands that wmake executes, including silent ones.
- The w32api project has been updated to the latest version.
- The os2api project has been enhanced - added multimedia headers and libraries and numerous fixes have been made to the header files.
- The debugger now supports the F7 key as a shortcut for "run to cursor". This is consistent with CodeView.
- New internal variable dbg\$ntid (next thread id) has been added to the debugger. This permits automated iteration of all threads.
- The wsampl tool has been updated to dynamically allocate storage for command line, where previously the command line length was limited to 128 characters.
- The FORTRAN compiler has been changed to preserve case of symbols with linkage other than FORTRAN. This is especially helpful when calling OS API functions and using case sensitive link step (now default).

6.13 Differences from Open Watcom Version 1.0

A number of problems has been corrected in Open Watcom C/C++ 1.1.

- Using the -ol option (loop optimization) sometimes resulted in generation of incorrect code.
- The *printf()* function could access too much data when formatting strings, causing page faults.
- NaNs, INFs and denormals were not handled correctly by the math emulation library.
- The assembler did not generate implicit segment override prefixes when accessing data in code segment.

- The *clock()* function sometimes produced incorrect results on OS/2 if the `TZ` environment variable was set.
- The Open Watcom editor (vi/viw) has been changed to store temporary files in directory designated by the `TMP` environment variable, instead of using the `TMPDIR` environment variable.
- Many packaging problems (missing files) have been resolved.

6.14 Differences from Version 11.0

Open Watcom C/C++ 1.0 is not substantially different from Watcom C/C++ version 11.0. There are however several changes that may require you to change your source and/or makefiles or project files.

- The C compiler now implements stricter function prototype processing. This may result in new warning or error messages on code that was previously compiling cleanly but contained mismatches between function prototypes and actual definitions. The C++ compiler did not have this problem.
- The linker now defaults to case sensitive symbol resolution. In most cases this is not a problem and may in fact avoid certain problems. If your project relies on case insensitive linking, please add *OPTION NOCASEEX* to your linker directives.
- Default windowing support has been removed from the runtime libraries.
- Components of the Win32 SDK and OS/2 Toolkits are no longer supplied.
- MFC libraries, source code and documentation are no longer supplied.

6.15 Changes in 11.0 that may Require Recompilation

Do not attempt to mix object code generated by earlier versions of the compilers with object code generated by this release or with the libraries provided in this release.

A new C++ object model has been implemented. If you have undefined references to `__wcpp_3_*` names, you have old object code. If you have undefined references to `__wcpp_4_*`, you have old libraries and new object code.

clock() The *clock* function accuracy has changed from 100 ticks per second to 1000 ticks per second (i.e., *CLOCKS_PER_SEC* has changed). Source code that uses the *clock* function and *CLOCKS_PER_SEC* in its calculations must be recompiled before linking the application with new libraries.

6.16 Major Differences from Version 10.6

The following sections summarize the major differences from the previous release.

- In general, we have improved Microsoft compatibility in our compilers (more warnings instead of errors, support for MS extensions, etc.) and tools.
- Some of the Win32 and 32-bit OS/2 versions of our tools are now available in DLL form.

EXE	DLL	Description
-----	-----	-----
wcc	wccd	16-bit x86 C compiler
wcc386	wccd386	32-bit x86 C compiler
wpp	wppdi86	16-bit x86 C++ compiler
wpp386	wppd386	32-bit x86 C++ compiler
wlink	wlink	Open Watcom Linker
wlib	wlibd	Open Watcom Library Manager

This provides better performance when using the Integrated Development Environment or Open Watcom Make. See the description of the `!loaddll` preprocessing directive in Open Watcom Make for more information.

Changes to the C++ Compiler for 11.0

- The C++ compiler now optimizes empty base-classes to occupy zero storage in the derived class memory layout. The C++ Working Paper recently allowed this optimization to be performed by conforming implementations. Furthermore, the optimization has speed and size benefits. There are certain classes of (broken) C++ programs that may not function properly with the new optimization. If you explicitly `memset()` an empty base class, you will be clearing memory that you may not expect to be cleared since the "zero sized" base class in actual fact shares storage with the first member of the derived class. A `memset()` of the entire derived class is fine though.
- We have added support for the `mutable` keyword which is used to indicate data members that can be modified even if you have a `const` pointer to the class.

Example:

```
class S {
    mutable int x;
    void foo() const;
};

void S::foo() const {
    x = 1;  // OK since it is mutable
}
```

- We have added support for the `bool` type along with `true` and `false`.
- We have added support for the `explicit` attribute. It marks a constructor so that it will not be considered for overloading during implicit conversions.

Example:

```
struct S {
    explicit S( int );
};

S v = 1;  // error; cannot convert 'int' to 'S'
```

Suppose the class was changed as follows:

Example:

```
struct S {
    explicit S(int );
    S( char );
};

S v = 1;  // OK; S( char ) is called
```

The fact that `S(int)` is not considered leaves `S(char)` as the only way to satisfy the implicit conversion.

- We have added support for name spaces.

```
namespace x {
    // anything that can go in file-scope
}
namespace {
    // anything in here is local to your module!
}
```

In the above example, you can access names in the namespace "x" by "x::" scoping. Alternatively, you can use the "using namespace x" statement (thereby eliminating the need for "x::" scoping). You can include a part of the namespace into the current scope with the "using x::member" statement. (also eliminating the need for "x::" scoping).

1. Name spaces eliminate the hand mangling of names. For example, instead of prefixing names with a distinguishing string like "XPQ_" (e.g., XPQ_Lookup), you can put the names in a namespace called "XPQ".
2. Name spaces allow for private names in a module. This is most useful for types which are used in a single module.
3. Name spaces encourage the meaningful classification of implementation components. For example, code-generation components might reside in a namespace called "CodeGen".

- We have added support for RTTI (Run-Time Type Information).
- We have added support for the new C++ cast notation. It allows you to use less powerful casts than the all powerful C-style cast and to write more meaningful code. The idea is to eliminate explicit casts by using a more meaningful new-style cast. The new C++ casts are:

```
reinterpret_cast < type-id >(expr)
const_cast < type-id >( expr )
static_cast < type-id >( expr )
dynamic_cast < type-id >( expr ) (part of RTTI)
```

- We have improved (faster) pre-compiled header support.
- We have added "long long" (64-bit integer) support in the form of a new `__int64` type.
- The default structure packing was changed from "zp1" to "zp2" in the 16-bit compiler and from "zp1" to "zp8" in the 32-bit compiler.
- The default type of debugging information that is included in object files is "Dwarf". It used to be "Watcom".

- A new double-byte string processing option has been added (zkl). When this option is specified, the local or current code page character set is used to decide if the compiler should process strings as if they might contain double-byte characters.

Changes to the C Compiler for 11.0

- We have improved (faster) pre-compiled header support.
- We have added "long long" (64-bit integer) support in the form of a new `__int64` type.
- The default structure packing was changed from "zp1" to "zp2" in the 16-bit compiler and from "zp1" to "zp8" in the 32-bit compiler.
- The default type of debugging information that is included in object files is "Dwarf". It used to be "Watcom".
- A new double-byte string processing option has been added (zkl). When this option is specified, the local or current code page character set is used to decide if the compiler should process strings as if they might contain double-byte characters.

Changes to the Code Generator for 11.0

- We support Microsoft-compatible in-line assembly formats using the `"_asm"` keyword.
- A new optimization, "branch prediction", has been added. This optimization is enabled by the "ob" or "ox" compiler options. The code generator tries to increase the density of cache use by predicting branches based upon heuristics (this optimization is especially important for Intel's Pentium Pro).
- We have added Multi-media Extensions (MMX) support to the in-line assemblers.
- We have added "long long" (64-bit integer) support in the form of a new `__int64` type.

Changes to the Compiler Tools for 11.0

- The Open Watcom Linker supports incremental linking.
- The Open Watcom Linker can now process COFF and ELF format object files, as well as OMF et al. The Open Watcom Linker can now read both AR-format (Microsoft compatible) libraries and old-style OMF libraries.
- Support for creating 16-bit DOS overlaid executables has been removed from the linker.
- The Open Watcom Library Manager (WLIB) can now process COFF and ELF format object files, as well as OMF et al. The Open Watcom Library Manager can now read/write both AR-format (Microsoft compatible) libraries and old-style OMF libraries. The default output format is AR-format and this can be changed by switches. The Open Watcom Library Manager can output various format import libraries.
- We have added Multi-media Extensions (MMX) support to the Open Watcom Assembler (WASM).

- A new version of the Open Watcom Disassembler (WDIS) can disassemble Intel or Alpha AXP object code files. It can process ELF, COFF or OMF object files and ELF, COFF or PE format (Win32) executables. The Open Watcom Disassembler looks at image file being input to determine the processor-type (defaults to Intel).

The old disassembler (WDISASM) has been retired and is not included in the package.

- We have added new tool front-ends that emulate Microsoft tools. These are:

- nmake
- cl
- link
- lib
- rc
- cvtres

These programs take the usual Microsoft arguments and translate them, where possible, into equivalent Open Watcom arguments and spawn the equivalent Open Watcom tools.

- Open Watcom Make now processes Microsoft format makefiles when the "ms" option is used.

Changes to the C/C++ Libraries for 11.0

- We have added multi-byte and wide character (including UNICODE) support to the libraries.
- We include run-time DLLs for the C, Math and C++ Libraries.
- We have added Multi-media Extensions (MMX) support to the libraries.
- The following new functions were added to the library...

multi-byte functions

- The **clock** function accuracy has changed from 100 ticks per second to 1000 ticks per second (i.e., **CLOCKS_PER_SEC** has changed).
- A "commit" flag ("c") was added to the `fopen()` *mode* argument.
- The global translation mode flag default is "text" unless you explicitly link your program with `BINMODE.OBJ`.
- Processing of the "0" flag in the format string for the `printf()` family of functions has been corrected such that when a precision is specified, the "0" flag is ignored.

Example:

```
printf( "%09.31f\n", 1.34242 ); // "0" flag is ignored
printf( "%09lf\n", 1.34242 ); // "0" flag is not ignored
```

- Support for printing `__int64` values was added to **printf** and related functions.
- Support for scanning `__int64` values was added to **scanf** and related functions.
- The Win32 **_osver** variable was added to the library.

- The Win32 *_winmajor*, *_winminor* and *_winver* variables were added to the library.

Changes to the DOS Graphics Library for 11.0

- The graphics library now performs the VESA test *before* testing for vendor specific graphics cards. This fix is intended to broaden the number of graphics cards that are supported.

Changes in Microsoft Foundation Classes Support for 11.0

- Version 4.1 of the 32-bit MFC is included in the package.
- Version 2.52b of the 16-bit MFC is included in the package.

Changes in Microsoft Win32 SDK Support for 11.0

- The Win32 SDK is supported for Windows 95 and Windows NT platforms.

Changes in Blue Sky's Visual Programmer for 11.0

- A new 32-bit version of Visual Programmer is included in the package. This version runs on 32-bit Windows 95 and NT. The 16-bit version of Visual Programmer is no longer included in the package.
- You can generate 16-bit applications with it, but you must be careful to avoid using Win95 controls.
- This new version fixes all known bugs in the previous version.

6.17 Changes in 10.6 that may Require Recompile

<i>_diskfree_t</i>	The struct members of the <i>_diskfree_t</i> structure has been changed from UNSIGNED SHORTs to UNSIGNED INTs. This is to deal with possible HPFS partitions whose size will overflow a short, as well as Microsoft compatibility.
<i>clock()</i>	The <i>clock</i> function accuracy has changed from 100 ticks per second to 1000 ticks per second (i.e., <i>CLOCKS_PER_SEC</i> has changed). Source code that uses the <i>clock</i> function and <i>CLOCKS_PER_SEC</i> in its calculations must be recompiled before linking the application with new libraries.

6.18 Major Differences from Version 10.5

The following sections summarize the major differences from the previous release of Watcom C/C++.

Windows 95 Help File Format

We have included Windows 95 format help files.

Changes to the C++ Compiler in 10.6

We have improved Microsoft compatibility so that Win32 SDK and MFC header files can be compiled without change. The following changes were required to support Win32 SDK header files.

- We recognize the single underscore versions of `__stdcall`, `__inline`, and `__fastcall` keywords.
- The `_fastcall` and `__fastcall` keywords are scanned but ignored since they refer to a particular Microsoft code generation technique. Open Watcom's generated code is always "fast".

The following changes were required to support MFC source code.

- When `/bt=DOS` is specified, define `_DOS`.
- When `/bt=WINDOWS` is specified, define `_WINDOWS`.
- When `/m[s|m|c|l|h]` is specified, define `__SW_M[S|M|C|L|H]` and `_M_I86[S|M|C|L|H]M`.

The compiler now supports the C++ Standard Template Library (STL). This library is available at the ftp site "butler.hpl.hp.com". When compiling applications that use the STL, you must use the "hd" compiler option for debugging info (the "hw" option causes too much debug information to be generated).

Changes to the C Compiler in 10.6

We have improved Microsoft compatibility so that Win32 SDK and MFC header files can be compiled without change. The following changes were required to support Win32 SDK header files.

- Support for the single underscore version of the `__stdcall` keyword.
- When `/bt=DOS` is specified, define `_DOS`.
- When `/bt=WINDOWS` is specified, define `_WINDOWS`.

The following changes were required to support SDK sample code.

- You can specify calling convention information in a function prototype and you do not have to specify the same information in the definition. (Note: This is required by the OS/2 Warp SDK samples.)
- Structured exception handling is supported (`__try`, `__except` and `__finally` keywords).
- Allow initialization of automatic array/struct data using variables and function calls.

Changes to the C Library in 10.6

The following new functions were added to the library.

`_getw` read int from stream file

`_putw` write int to stream file

The *clock* function accuracy has changed from 100 ticks per second to 1000 ticks per second (i.e., *CLOCKS_PER_SEC* has changed).

Changes in Microsoft Foundation Classes Support for 10.6

- Version 3.2 of the 32-bit MFC is included in the package.
- Version 2.52b of the 16-bit MFC is included in the package.

Changes to the Image Editor in 10.6

- Support has been added for 256 color bitmaps.
- Support has been added for 16 X 16 icons.
- Support has been added for 48 X 48 icons.

Changes to the Dialog Editor in 10.6

- Support has been added for Windows 95 controls.
- Support has been added for adding new control styles to existing controls.
- Support has been added for new dialog styles.
- Support has been added for allowing help IDs to be specified in dialog and control statements.
- Support has been added for generating new resource statements in .RC files.

Changes to the Resource Editor in 10.6

- Support has been added for new Windows 95 DIALOGEX resource type.
- Support has been added for generating new DIALOGEX resource statements in .RC files.

Changes to the Resource Compiler in 10.6

- Support has been added for extended styles for dialogs.
- Support has been added for the RCINCLUDE keyword.

6.19 Major Differences from Version 10.0

- New installation program
- Visual Programmer for Windows (MFC) applications
- MFC 3.0 support
- Native C++ exception handling support
- Improved language compatibility with Microsoft
- Browser can now be used to browse C code
- OS/2 3.0 Warp support
- Toolkit for OS/2 1.3
- Windows NT 3.5 support
- Toolkit for Windows NT 3.5
- Windows 95 (Chicago) support
- Source Revision Control System hooks in editor
- TCP/IP remote debug servers for OS/2 and Windows NT/95

In addition to these new features, we have also made a number of improvements to the software.

1. The editor is more tightly integrated with the IDE.
2. It is now easier to select your own favorite editor from the IDE.
3. The keyboard interface in the Integrated Development Environment (IDE) has been improved.
4. The "fr" option, which is supported by the compilers & assembler, can be used to name the error file drive, path, file name and/or extension.
5. We have added the "t<number>" option to the C++ compiler to set the number of spaces in a tab stop (for column numbers in error messages).
6. The C compiler now supports @filename on the command line like the C++ compiler currently does.

7. The "__stdcall" linkage convention has changed. All C symbols (extern "C" symbols in C++) are now suffixed by "@nnn" where "nnn" is the sum of the argument sizes (each size is rounded up to a multiple of 4 bytes so that char and short are size 4). When the argument list contains "...", the "@nnn" suffix is omitted. This was done for compatibility with Microsoft. Use the "zz" option for backwards compatibility.
8. The 32-bit "__cdecl" linkage convention has changed. Watcom C/C++ 10.0 __cdecl did not match the Microsoft Visual C++ __cdecl in terms of the binary calling convention; Visual C++ saves EBX in a __cdecl function but Watcom C/C++ 10.0 modified EBX. Watcom C/C++ has been changed to match Visual C/C++.

If you wrote a "__cdecl" function in an earlier version of Watcom C/C++, the EBX register was not saved/restored. Starting with release 10.5, the EBX register will be saved/restored in the prologue/epilogue of a "__cdecl" function.

Another implication of this change is that "__cdecl" functions compiled with an earlier version of Watcom C/C++ don't match the calling conventions of the current version. The solution is either to recompile the functions or to define a "__cdecl_old" pragma that matches the old calling conventions.

```
#pragma aux __cdecl_old "_" \
    parm caller [] \
    value struct float struct routine [eax] \
    modify [eax ebx ecx edx];

#pragma aux (__cdecl_old) foo;

extern int foo( int a, int b );

void main()
{
    printf( "%d\n", foo( 1, 2 ) );
}
```

9. We now allow:

```
extern "C" int __cdecl x;
```

It must be extern "C" for __cdecl to take effect since variables have their type mangled into the name for "C++" linkage.

10. In C++, we have removed the warning for "always true/false" expressions if the sub-expressions are constant values.
11. We have added support for:

```
#pragma pack(push,4);
#pragma pack(push);
#pragma pack(pop)
```

12. We have added support for:

```
#pragma comment(lib, "mylib.lib")
```

which has the same semantics as:

```
#pragma library( "mylib.lib" )
```

13. We have added support for expanding macros in the code_seg/data_seg pragmas:

```
#define DATA_SEG_NAME "MYDATA"
#define CODE_SEG_NAME "MYCODE"

#pragma data_seg( DATA_SEG_NAME )
int x = 3;

#pragma code_seg( CODE_SEG_NAME )
int fn() {
    return x;
}
```

14. We have fixed the 16-bit compiler so that it matches the Microsoft 16-bit C compiler for the following cases:

- If a pascal function is defined when compiling for Windows 3.x, use the fat Windows 3.x prologue in the function.
- If a cdecl function is defined when compiling for Windows 3.x, use the fat Windows 3.x prologue in the function.

15. We have fixed the compiler so that

```
#include </dir/file.h>
```

works as expected (it was searching along the INCLUDE path only).

16. In C++, we have fixed a problem where an import was generated in the object file for a virtual function call. This will reduce the size of executables under certain circumstances.

17. In C++, we have removed the prohibition of pointer to array of unknown size declarations.

Example:

```
int (*p) [];
```

18. In C++, we have fixed the diagnosis of lexical problems during macro expansion to remove spurious warnings.

Example:

```
#define stringize( x )  #x

stringize( 2131231236172637126371273612763612731 )
```

19. We have corrected the check for too many bytes in #pragma for assembler style aux #pragmas.

20. Undeclared class names in elaborated class specifiers are now declared in the nearest enclosing non-class scope. Undeclared classes are also allowed in arguments now.

Example:

```
struct S {
    // used to declared ::S::N but now declares ::N
    struct N *p;
};

void foo( struct Z *p );    // declares ::Z
```

21. We have fixed unduly harsh restriction on virtual ...-style functions. They are now allowed in single inheritance hierarchies as long as the return type is not changed when the virtual function is overridden. In multiple inheritance hierarchies, an implementation restriction is still present for generating a 'this' adjustment thunk for virtual ...-style functions.
22. We have fixed line number information for multi-line statement expressions in some weird cases.
23. We have fixed function template parsing of user-defined conversions that use an uninstantiated class in their operator name.

Example:

```
void ack( int );

template <class T>
struct S {
    S( T x )
    {
        ack( x );
    }
};

template <class T>
struct W {
    operator S<T>();
};

template <class T>
W<T>::operator S<T>() {
    return 0;
}
```

24. We have fixed a compiler problem that caused a linker warning "lazy reference for <virtual-fn> has different default resolutions" in cases where the compiler or programmer optimized virtual function calls to direct calls in modules that also contained virtual calls.

Example:

```
T.H
struct S {
    virtual int foo() { return __LINE__; }
};
struct T : S {
    virtual int foo() { return __LINE__; }
};
```



```
T1.CPP
#include "t.h"
struct Q : T {
    virtual int foo() { return S::foo() + __LINE__; }
};

void foo( T *p )
{
    Q y;
    y.foo();
    p->foo();
}
```

```
T2.CPP
#include "t.h"

void foo( T *p );

void ack( T *p ) {
    p->foo();
    foo(p);
}

main() {
    T q;
    ack( &q );
}
```

25. When a class value is returned and is immediately (in the same expression) used to call a member function, the value may not be stored in memory.

Work around: introduce a temporary

Example:

```
struct S {
    int v;
    int member();
};

S foo();

void example( void )
{
    // foo().member(); // replace this line with:
    S temp = foo();
    temp.member();
}
```

26. Throwing pointers to functions did not work when the size of a function pointer is greater than the size of a data pointer.

Work around: place the function pointer in a class and throw the class object.

27. We have fixed default argument processing for const references to an abstract class. The following example would not compile properly:

Example:

```
struct A {
    virtual int foo() = 0;
};

A &foo();

void ack( A const &r = foo() );

void bar() {
    ack();
}
```

- 28. We have made "DllMain" default to extern "C" linkage for Microsoft Visual C++ compatibility.
- 29. We have duplicated a Microsoft Visual C++ extension that was required to parse the Windows 95 SDK header files.

Example:

```
typedef struct S {
} S, const *CSP;
^^^^^ not allowed in ISO C or ISO C++
```

- 30. We now do not warn about starting a nested comment if the comment is just about to end.

We also fixed the code that figures out where a comment was started so that a nested comment warning is more helpful.

Example:

```
/*//////////*/
  ^_
```

- 31. We have fixed a problem where extra informational notes were not being printed for the error message that exceeded the error message limit.

Example:

```
// compile -e2
struct S {
    void foo();
};

void foo( S const *p )
{
    p->foo();
    p->foo();
    p->foo();
    p->foo();
}
```

- 32. We have fixed a problem where the line number for an error message was incorrect.

Example:

```
struct S {
    void foo() const;
    void bar();
};

void S::foo() const
{
    bar();

    this->bar();
}
```

- 33. We have fixed output of browser information for instantiated function template typedefs.
- 34. We have upgraded the C++ parser so that casts and member pointer dereferences can appear on the left hand side of the assignment expression without parentheses.

Example:

```
p->*mp = 1;
(int&)x = 1;
```

- 35. In several cases, when a function return or a construction was immediately dotted in an expression, the generated code was incorrect:

Example:

```
struct S {
    int x;
    int foo();
};

extern S gorf();

void bar()
{
    gorf().foo();
}
```

The work around was to break the statement in two:

Example:

```
S temp = gorf();
temp.foo();
```

- 36. In several cases, when a function return or a construction was immediately addressed in an expression, the generated code was incorrect:

Example:

```
struct S {
    int x;
};

extern void fun( S* );

extern S gorf();

void bar()
{
    fun( &gorf() );
}
```

The work around was to break the statement in two:

Example:

```
S temp = gorf();
fun( &temp );
```

37. We have added support for:

```
#pragma error "error message"
```

Use the ISO/ANSI method because it is more portable and acceptable (Microsoft header files use the less portable #pragma when there is a perfectly fine, portable way to issue a message).

The portable, acceptable method is:

```
#error "error message"
```

38. We have added support for `__declspec(dllexport)`, `__declspec(dllimport)`, `__declspec(thread)`, and `__declspec(naked)` for Win32 (i.e., WinNT 3.5 and Win95) programs. Here are some examples:

Example:

```
__declspec(dllexport) int a;           // export 'a' variable
__declspec(dllexport) int b()          // export 'b' function
{
}

struct __declspec(dllexport) S {
    static int a;                       // export 'a' static
    member
    void b();                           // export 'b' member fn
};

extern __declspec(dllimport) int a;     // import 'a' from a .DLL
extern __declspec(dllimport) int b();  // import 'b' from a .DLL

struct __declspec(dllimport) I {
    static int a;                       // import 'a' static
    member
    void b();                           // import 'b' member fn
};
```

39. The C++ compiler generates better error messages for in-class initializations and pure virtual functions.

Example:

```
struct S {
    static int const a = 0;
    static int const b = 1;
    void foo() = 0;
    void bar() = 1;
    virtual void ack() = 0;
    virtual void sam() = 1;
};
```

40. We have fixed macro processing code so that the following program compiles correctly. The compiler was not treating "catch" as a keyword after the expansion of "catch_all".

Example:

```
#define catch(n) catch(n &exception)
#define xall (...)
#define catch_all catch xall

main()
{
    try{
    }
    catch_all{
    }
}
```

41. We have fixed a problem where `#pragma code_seg` caused a page fault in the compiler when the `code_seg` was empty.
42. We have fixed a rare problem where a `#include` of a file that was previously included caused the primary source file to finish up if the CR/LF pair for the line that the `#include` was on, straddled the C++ compiler's internal buffering boundary.
43. We have added support for `#pragma message( "message text" )`. It outputs a message to stdout when encountered. It is used in Microsoft SDK header files to warn about directly including header files and obsolete files.
44. We have fixed `#pragma code_seg/data_seg` to properly set the class name of the new segment in the object file.
45. We have fixed a problem with the `-zm -d2` options that caused a compiler fault in some circumstances.
46. We have fixed default library records in .OBJ file so that user libraries are ahead of default compiler libraries in the linker search order.
47. We have fixed handling of intrinsic math functions so that the code generator will treat functions like `sqrt` as an operator.
48. We have added support for using OS-specific exception handling mechanisms for C++ exception handling during code generation. Enable it with the new `-zo` option.

49. `__stdcall` functions now have Microsoft Visual C/C++ compatible name mangling.
50. We have added a number of new functions to the C Library. These have been added to improve Microsoft compatibility.

```
dllmain (nt only)
libmain (nt only)
_access
_dos_commit
_dup
_ecvt
_fcvt
_fstat
_fstrdup
_gcvt
_itoa
_itoa
_locking
_lseek
_ltoa
_ltoa
_memicmp
_set_new_handler
_stat
_strdate
_strdup
_stricmp
_strlwr
_strnicmp
_strrev
_strtime
_strupr
_tolower
_toupper
__isascii
__iscsym
__iscsymf
```

51. In version 9.5, the linker used to include LIBFILE object files in reverse order (i.e., the last one listed was the first to be included). We have corrected this behavior so that they are included in the order listed.

Directive		Old Order	New Order
-----		-----	-----
FILE	obj_a	3	3
LIBFILE	obj_b	2	1
LIBFILE	obj_c	1	2
FILE	obj_d	4	4

In the above example, the object files will be included in the order indicated (LIBFILE object files are always included first).

Changes in 10.5 that may Require Recompilation

- __stdcall** If you use the `__stdcall` attribute within a program then you must re-compile the function definition and all callers of the `__stdcall` function.
- __cdecl** The `__cdecl` attribute is not heavily used in Win32 programming so the impact should be minimal but if you do use `__cdecl` within your own programs, a re-compilation will be necessary.

6.20 Major Differences from Version 10.0 LA

If you have .tgt files created with the Limited Availability or Beta Integrated Development Environment, when you load them, the target window may say "Obsolete Form: rename target type". If it does:

1. Select the target window by clicking in it,
2. Choose "rename target" from the target menu (a rename target dialog will appear),
3. Reselect the target type for this target (e.g., Win32 EXE), and
4. Select OK.

You should not continue to use .cfg files from the Limited Availability version of the compiler. Several new features have been added. Using the old files will cause problems.

The C++ compiler calling conventions have changed. Any program that passes a "data only" class or struct as a parameter, or returns a C++ object will need to be recompiled. We recommend that you recompile your application.

The C++ compiler now supports the use of the `__export`, `__cdecl`, `__pascal`, `__stdcall` and `__syscall` keyword on class definitions. These keywords will affect all members of the defined class.

6.21 Major Differences from Watcom C9.5 /386

- The functionality of Watcom C/C++(16) and Watcom C/C++(32) is included in a single package.
- An Integrated Development Environment for Windows 3.x, Windows NT, Windows 95 and OS/2 PM is included.
- New, redesigned debugger with GUI interfaces for Windows 3.x, Windows NT, Windows 95 and OS/2 PM is included.
- The optimizer has been enhanced.
- C++ Class Browser
- New, redesigned user interface for the Profiler.
- New support for C and C++ precompiled header files.
- Windows resource editing tools are included:

Dialog Editor
Bitmap Editor
Resource Editor
Menu Editor
String Editor
Accelerator Editor

- Windows development tools are included:

Dr. Watcom (a post mortem debug utility)
Spy (Windows message spy program)
DDESpy
Heap Walker
Zoom

- On-line documentation is included.
- Microsoft Foundation Classes for 32-bit Windows applications (MFC 4.1) and 16-bit Windows 3.1 applications (MFC 2.52b) is included.
- Creation of FlashTek DOS extender applications is supported.
- Compiler executables have been created that run under all supported operating systems. They are located in the BINW directory.

Items No Longer Supported

- PenPoint development
- Debugging of Ergo OS/386 DOS extender applications
- DESQView remote debugging

Changes in 10.0 that may Require Recompilation

All C++ applications will require recompilation due to changes in the underlying object model. C applications should not require recompilation, but you should recompile your application if you want to take full advantage new features in the debugger. The changes to the C++ object model are:

- Virtual table layout changed (NULL entry at offset 0 removed)
- derived class packing adjusted to minimize padding bytes
- exception handling code is improved (incompatible with 9.5)
- name mangling for 'char' reduced from two chars to one char

6.22 Major Differences from Watcom C9.01 /386

- C++ support added
- Pentium optimizations added ("5t", "5s", "fp5" options)
- Windows NT support added
- Microsoft resource compiler (RC.EXE) replaced with Watcom resource compiler (WRC.EXE)
- OS/2 libraries modified so that single library supports single and multiple threads and DLL's
- "fpi287" switch renamed to "fp2"
- #pragma intrinsic and #pragma function added
- 80x87 usage modified so that compiler assumes all eight registers are available upon entry to function instead of only four registers. "fpr" option added for reverse compatibility with object files compiled with previous versions of compiler

Changes that may Require Recompilation

The *stat* structure in "stat.h" changed in version 9.5. Any object files compiled with an earlier version of the compiler will have to be recompiled if they use the *stat* structure.

A new function *_grow_handles* was added to version 9.5 for growing the number of available file handles. Details on how to use this function can be found in the *Watcom C/C++³² Commonly Asked Questions & Answers* booklet.

If you compile with structure packing (/zp2, /zp4, /zp8) or use the "pack" pragma to specify a packing value other than 1, and you have structures that contain other structures, field offsets and structure sizes may be different depending on the contents of the structures.

In version 9.01, the new Windows Supervisor now has 32 user-defined callbacks. If you have any user-defined callbacks in your Windows program, you must recompile because the constant definitions have changed.

In version 9.0, the compiler will not use FS by default to avoid conflicts with new operating systems (OS/2 2.0, PenPoint, Windows NT). This will cause compile errors if you have defined a pragma that uses the FS register.

In version 8.5, the compiler was changed so that it by default does not save and restore segment registers across function calls. This is to solve problems that occur where a segment register is saved and restored in a function that tries to free the segment. When the segment register is popped from the stack in the epilogue, a general protection exception occurs because the selector is no longer valid. In order to provide backward compatibility with existing code, we have added a "-r" option that will cause the compiler to save and restore segment registers across calls. The C run-time library has been compiled with the "-r" option so that it will work should you choose to compile your application with the same option.

The packing algorithm was also changed in version 8.5. If you are using one of the "-zp2, -zp4, or -zp8" options to pack structures, you must recompile your application. The packing algorithm has been changed

so that the minimum number of slack bytes are inserted into structures to align fields onto their required alignment.

6.23 Major Differences from Watcom C9.0 /386

- Windows 3.1 SDK components
- Support for Windows 3.1 DLLs
- On-line Watcom C Library Reference help file for OS/2 2.0. Help can be accessed by issuing the command "VIEW WATCOMC". The command can be followed by a topic. e.g. VIEW WATCOMC PRINTF.

Command Line Options added to Watcom C9.0 /386

<i>4r</i>	Use register calling conventions and optimize for 486.
<i>4s</i>	Use stack calling conventions and optimize for 486.
<i>ee</i>	Generate a call to __EPI at the end of a function
<i>ep{=number}</i>	Generate a call to __PRO at the start of a function
<i>oe</i>	In-line user defined functions.
<i>or</i>	Reorder instructions to take advantage of 486 pipelining.
<i>zff</i>	Allows the FS register to be used by the code generator for far pointers.
<i>zfp</i>	Disallows use of the FS register. This is the default in flat memory model, because operating systems are now using FS to point to important information. (e.g. OS/2 2.0, PenPoint, Windows NT).
<i>zm</i>	Places each function into a separate segment. This will allow for smart linking.

6.24 Major Differences from Watcom C8.5 /386

Several major items have been added to Watcom C9.0 /386. They are:

- 486 instruction selection
- 486 instruction scheduling
- Tail recursion elimination
- Function inlining
- strcmp function optimized for 486
- Support for OS/2 2.0
- New keywords added (_far16, _Seg16, _Packed, _Cdecl, _Pascal, _System)
- Linkage pragma for compatibility with IBM C Set/2
- Based pointers
- Machine language code sequences can now be entered into pragmas using assembly language instead of numeric codes.

- Remote debugging over the parallel port using either a "LapLink" cable or a "Flying Dutchman" cable.
- Remote debugging of PenPoint applications

Command Line Options added to Watcom C8.5 /386

<i>dl+</i>	to generate line number information plus typing information for global symbols and local structs and arrays
<i>ei</i>	force all enumerated types to be of type 'int'
<i>en</i>	emit routine name before prologue
<i>ez</i>	generate PharLap EZ-OMF object file
<i>fpi287</i>	for programs that run on a system with a 287
<i>of</i>	to generate traceable stack frames
<i>of+</i>	to generate traceable stack frames for all functions
<i>om</i>	to generate in-line math functions
<i>p</i>	to generate preprocessor output
<i>pl</i>	to generate preprocessor output with #line directives
<i>pc</i>	to generate preprocessor output preserving comments
<i>plc</i>	to generate preprocessor output with #line directives and preserving comments
<i>r</i>	save/restore segment registers across calls
<i>we</i>	treat warnings as errors

6.25 Major Differences from Watcom C8.0 /386

Several major items were added to Watcom C8.5 /386. They are:

- Royalty-free 32-bit DOS extender
- Windows 3.0 support
- Windows SDK components included
- Improved optimizations
- OS/2 hosted version of the compiler
- The compiler now generates Microsoft format object files by default. Use the '/ez' option to generate Phar Lap EZ-OMF object files.
- More library functions to create higher compatibility with Microsoft C 5.1 and Microsoft C 6.0.
- Preprocessor output from the compiler
- Standalone help utility
- Object module convert utility can convert debugging information into CodeView format
- Protected-mode version of the linker
- Debugger support for ADS applications
- Support for Pharlap 3.0 and Pharlap 4.0
- Support for Tenberry Software DOS/4G
- Support for Intel 386/486 Code Builder Kit
- Support for UNICODE. Literal strings and character constants that are preceded by "L" map code page 437 onto UNICODE unless one of the options "-zk0, -zk1, or -zk2" is specified.

Command Line Options added to Watcom C8.0 /386

<i>d2</i>	symbolic debugging information
<i>oc</i>	disable "call" followed by "ret" being changed into a "jmp" optimization
<i>u<name></i>	undefine a pre-defined name
<i>zc</i>	places literal strings in the CODE segment
<i>zk{0,1,2}</i>	double-byte character support

6.26 Major Differences from Watcom C7.0 /386

Several major items have been added to Watcom C8.0 /386. They are:

- Linker
- Librarian
- Graphics library
- More library functions to create higher compatibility with Microsoft C 5.1 and Microsoft C 6.0.

The professional edition also gives you the following most asked for features:

- Protected-mode version of the compiler
- Full-screen source-level debugger
- Execution profiler

Protected-mode Compiler and Linker

The protected-mode version of the compiler "WCC386P.EXE" and linker "WLINKP.EXE" use a DPMI compliant DOS extender. This allows you to run the compiler and linker on a normal DOS system or in a Windows 3.x DOS box operating in enhanced mode.

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EXHIBIT A.

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8 Trouble-Shooting

This section is intended to provide you with help on some of the common (and not so common) problems that users have encountered when trying to run the software.

Symptom: The message "Cannot connect to batch spawn server" is displayed.

Resolution: This message occurs when the Open Watcom IDE is trying to connect to the batch server. The batch server is a program that is employed by the Open Watcom IDE to run "makes" in the background. There are several reasons why you may receive this message.

- *Installing Under the Host System and Selecting "Modify System Files" During Install*

For the IDE to run properly under a particular host operating system, the install program must be run on that system.

It is very important to check the "Modify System Files" when running the install on the host platform. Otherwise, changes required in the host system's environment will not be made. If this is the problem, you should re-install under the host environment and select this option.

- *System stability*

If for some reason the operating system has become unstable, the IDE may have trouble connecting to the batch server. This can happen if you have run a badly behaved application that caused an exception, for example a page fault. If this is the problem, you should shutdown the operating system and reboot.

- *Running Windows 3.1 in Enhanced Mode*

The IDE requires at least version 3.1 of Windows to be running, in enhanced mode. Windows 3.0 is not supported.

Running Third-Party Shell Programs

If you run 4DOS.COM or some other shell besides COMMAND.COM, you must change the "BATCHBOX.PIF" file accordingly. You can do this using the standard Windows PIF editor.

Running Virus Checkers

If you are running Central Point's PCTOOLS, VWATCH, or some other virus checker, it may be conflicting with our batch server. To isolate the problem, uninstall the virus checker and see if the IDE works again.

386 Enhanced "Exclusive in Foreground" Option

If you go to the Windows Control Panel, "Scheduling Options" of "386 Enhanced", and the "Exclusive in Foreground" checkbox is checked, uncheck it. This causes our batch server to starve under Windows.

Changes to the SYSTEM.INI File

The WDEBUG.386 driver is required to be installed in the [386Enh] section of SYSTEM.INI. This should have been done automatically by the WATCOM install program when the software was installed under the host environment and the "modify system files" option was selected.

It has been reported that the line

```
OverlappedIO=on
```

not appear in the SYSTEM.INI file.

It appears the user's problem was related to conflicts with other devices installed in the SYSTEM.INI file. On its own, the above line does not appear to affect the execution of the IDE.

It has been reported that the line

```
NoEMMDriver=ON
```

not appear in the SYSTEM.INI file. It will prevent a link from succeeding in the IDE.

8.1 Win-OS/2 and OS/2 Specific

Installing for a Win-OS/2 Host

For Win-OS/2, the install must be run under OS/2 and "Modify System Files" must be selected. This installs the batch server under OS/2, which we require under Win-OS/2. Then the Windows install program should be run under Win-OS/2 so that the program groups will be created. "Modify System Files" should also be checked for the Win-OS/2 install.

Named Pipe Conflicts

Under OS/2, the batch server is a named pipe. Win-OS/2 uses the OS/2 named pipe for the batch server, hence the OS/2 IDE will conflict with the Win-OS/2 IDE if they are both running. If this is the problem, you should run only one of the OS/2 and Win-OS/2 IDE, not both at the same time.

3

386 Enhanced 10

4

4DOS 10

A

anti-virus 10

B

batch server 10, 94
 BATCHBOX.PIF 10
 benchmarking 37
 BINMODE 67

C

__cdecl 81
 class browser 2
 class libraries 3
 clock 63, 67-68, 70
 clock() 63
 CLOCKS_PER_SEC 63, 67-68, 70
 COMMAND.COM 10
 CONFIG.SYS 11
 Control Panel 10
 cross-platform 2

D

DDE spy 3
 debugger 11
 DELDIR environment variable 10
 DOS extender 3
 DPMI specification 5
 Dr. Watcom 3

E

editor 2
 Enhanced System Editor 10
 environment variables
 DELDIR 10
 PATH 8
 WATCOM 8
 EPM 10
 __export 81

F

fastest 16-bit code 37
 fastest 32-bit code 37

G

GUI tools 2

H

hardware requirements 7
 heap walker 3
 host platforms supported 1

I

IDE 10
Image Type 16-17
InDOSPolling 10
INSTALL 9
installation
 incremental 9
 modifications to files 9
 multiple operating systems 8
 previous version 7
installing Open Watcom C/C++ 7
 DOS 9
 OS/2 9
 Windows 3.x 9
 Windows 95/98/Me 9
 Windows NT/2000/XP 9
__int64 65-67

N

NMPBIND 11
NoEMMDriver 10, 94

O

OMF specification 5
OPTION NOCASEEX 63
_osver 67
OverlappedIO 10
overview 1

P

__pascal 81
PATH environment variable 8
performance analysis 2
PIF 10
platforms supported 1
printf 67

printf() 62
product overview 1
Program Information File 10

R

read-me file 4
resource compiler 3
resource editors 2

S

scanf 67
self-help 3
SETUP 9
software requirements 7
spy 3
stat 57
__stdcall 81
struct stat 57
__syscall 81
SYSTEM.INI 10, 94

T

Target Environment 16-17
target platforms supported 1
technical support 3-4

U

UNDELETE 10

W

WATCOM environment variable 8

WHELP 32

Windows

 386 Enhanced 10

 Control Panel 10

 InDOSPolling 10

 NoEMMDriver 10, 94

 OverlappedIO 10

 SYSTEM.INI 10, 94

_winmajor 68

_winminor 68

_winver 68

Z

zoom 3